



University of Applied Sciences

**Synthetic Data Generation for Skin Cancer
Detection: Addressing Bias and Accessibility in
Dermatological AI**

Background Research – Literature Review

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Abstract

The detection of skin cancer is a critical problem because all the popular dermatological datasets like ISIC, HAM10000, BCN20000, and Fitzpatrick17k have imbalanced representation of Fitzpatrick skin types and subgroups by sex. To overcome these differences, a synthetic data generation approach needs to be combined with semi-supervised learning (SSL) systems to enhance fairness without increasing the cost of annotation or compromising clinical reliability. It focuses on conditional generative adversarial networks (GANs) and diffusion models to generate synthetic dermatology image with a controlled skin-tone variability that ensures quality by detecting artifacts and hallucinations and avoiding amplifying bias. FixMatch and MixMatch are adaptations of SSL algorithms which include subgroup-aware pseudo-label filtering and dynamic mixing policies to balance the contribution of synthetic and real data, directed by measures of fairness such as stratified ROC-AUC, PR-AUC, sensitivity at constant specificity and calibration curves. The active learning tactics focus on specialized annotation of uncertain cases in the underrepresented groups with the aim of allocating resources efficiently. Some of the deployment considerations are model registry and version control systems which are used to monitor demographic distributions, synthetic data provenance, and performance drift that would maintain transparency and adherence to privacy. It uses domain adaptation and augmentation policies to simulate diverse acquisition conditions and overcome synthetic-to-real domain shift and cross-dataset generalization constraints. Fully automated validation procedures using locked and external test sets stratified based on the attributes that are under protection ensure fair diagnostic outcome. This combined framework will develop equitable, cost-effective, and clinically plausible AI systems to detect skin cancer in different patient groups.

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1. CONTEXT AND BACKGROUND

1.1 CLINICAL STAKES IN SKIN CANCER DETECTION

Skin cancer is still a significant burden to healthcare systems largely because early detection of the disease has a direct effect on the overall survival of the patient as well as the cost of treatment. Clinical interests in this field are increased by the range of different types of skin cancer basal cell carcinoma, squamous cell carcinoma, and melanoma, which demand individual diagnostics (Raju et al., 2024). False identification or delay in recognizing them can cause aggressive disease development which increases morbidity and resource utilization.

This supports the need to have correct, prompt, and equitable algorithms that can execute dependably in the presence of varied populations of patients. The existence of representation gaps in datasets increases these stakes. Most popular dermatology datasets including HAM10000, ISIC archives, BCN20000, and Fitzpatrick17k are not distributed equally and do not represent skin tone, sexes, and rare types of lesions. Biased datasets usually contain an overskewed sample of a lighter skin tone, which results in models being constructed under a biased model, which starts to work poorly on underrepresented populations (Pakzad et al., 2022). This is in reality translated into a clinical danger: lesions in skin with darker tones may be differentiated with lower sensitivity or not detected at all. Moreover, its reliance on unlabeled examples of rare conditions does not facilitate complete validation in such demographic subgroups.

The other cause of deployment realism is the labeling of quality. Manual annotation is still expensive and time-consuming because it usually involves expert dermatologists to differentiate only minor morphological features (Dhawan & Nijhawan, 2024). The control of data sharing is also restricted by the privacy restrictions posed by regulations such as GDPR or HIPAA. Alongside the large public datasets, erroneous images with the wrong label are presented, even non-clinical images are found in Fitzpatrick17k, which adversely affect the results of the training process (Pakzad et al., 2022). This highlights the importance of high-integrity annotation pipelines together with pre-systematic filtering of noise of datasets prior to incorporation into models to be used in real world clinical practice.

Algorithms fairness will be used to help reduce the risks of such gaps. Approaches like meta-balanced networks modify geometry in feature space instead of simply re-weighting loss terms to deal with differences between skin tone groups (Wang et al., 2022). These strategies are intended to produce greater congruency in decision limits at different demographics levels. Nevertheless, fairness interventions need to be implemented on the limitations of ensuring that the overall utility of the available diagnostic measures remains high in absolute terms; prioritizing balancing measures of subgroups to that of global performance can lead to the reduction in utility in the case of critical care.

The use of GANs to produce synthetic data also seems to help address the lack of datasets by adding more minority subgroups, without the annotation costs imposed being prohibitive (Georges-Filteau and Cirillo, 2020). However, one of the limitations of the domain shift between synthetic and real clinical images raises uncertainty regarding the generalization ability. Unless delicate texture patterns specific to genuine lesions are recreated with faithfulness, as is typical with models generating dermoscopic-like images, when used in practice, models trained on such large numbers of fake lesions are prone to false diagnoses of true ones. To deal with this question it is crucial to have tight cross domain assessment metrics like ROC-AUC and sensitivity to fixed specificity level to measure robustness on subsets of real patients.

Transparency in performance reporting is very critical in clinical trust. Deficiencies in subgroup performance measurements can be covered by merely high average accuracy. Regulatory norms are progressively exposing themselves towards intricate categorized assessments in guarded features such as the labels of sex and skin color (Sagers et al., 2023). Calibration curves and PR-AUC scores give a more insight into the correspondence between probability outputs and the probability of the actual outcomes, which is crucial in cases of situations where AI outputs is included in the decision-making process of clinicians; otherwise, it operates independently.

It becomes even more stakes when the environment is not the well-endowed hospitals. The areas of subspecialist shortage or absence require cost-effective solutions. The use of models, which apply the use of the SSL strategies like FixMatch or MixMatch might prove to be beneficial due to the capacity to utilize huge amounts of unsupplied data and yet maintain affordable computational costs. Nevertheless, risks of contamination occur when unlabeled pools have spurious correlations, similar methods to RSMATCH have been proposed to alleviate pseudo-label propagation error in training.

Workflow wise, integrating requires taking into consideration MLOps concepts on continuous monitoring after implementation. Clinical settings are suffering temporal drift: a device that collects lesion images, population demographics to which a clinic serves, or a season change in the number of cases of presentation can all change the input distributions in subtle ways (Kore et al., 2024). The systems should track performance and generate retraining based on updated data whenever drift is threatening the reliability of the model.

The strength of synthetics to real generalization, the integrity of labels, these when added together with representational fairness, and the deployability of the context, position the safety envelope within which dermatology AI systems must perform. Any one of them not thoroughly examined undermines patient outcomes and the level of trust that clinicians place in automated support tools (Raju et al., 2024).

1.2 LANDSCAPE OF DERMATOLOGY IMAGE DATASETS

1.2.1 ISIC SERIES AND THEIR EVOLUTION

The International Skin Imaging Collaboration (ISIC) series has gradually become one of the best used dermatology image repositories with many cutting-edge machine learning systems in skin cancer detection. Initially, ISIC was an aggregation platform for dermoscopic images from multiple international clinics and datasets such as HAM10000 were becoming hugely used benchmarks. HAM10000's creation was already a long project, with over 20 years of collecting and annotating images from geographically diverse centres, including Barcelona, New York City, Wollstonecraft, Brisbane, and Athens (Das, 2021). This broad timeline reflects the logistical and cost difficulties inherent in compiling good quality dermatological datasets. Even for the well-resourced collaborations, boosting representation amongst rare lesions or minority subgroups is slow and requires significant clinical coordination.

The corpus grew in size and on the variety of tasks with the successive iterations of the ISIC 2018, 2019 and 2020 challenges. However, analyses of these data sets show that there are persistent imbalances: Lighter skin colors dominate in all editions. Quantitative analysis with individual typology angle (ITA) measurements confirmed the findings that most images in ISIC 2018 were in the range of lighter pigmentation as measured by ITA. This distribution skew has a direct interaction with the fairness of models; classifiers trained on a majority of samples of a lighter skin tend to show a lower performance rate when it is applied to darker skinned patients.

The Fitzpatrick17K dataset was used in part to address this lack, and was intended to reach coverage of all six Fitzpatrick skin types with greater than 600 examples for each type. But such datasets integrated into the pipeline, side-by-side with ISIC archives, could allow researchers to focus on more equitable training distributions without the expense of data volume. The ISIC workshops also sparked off joint benchmarking to other collections like Dermfit Image Library, Diverse Dermatology Images (DDI) PAD-UFES-20, PH2. However, very few contain clear Fitzpatrick labels, which are important for quantitative subgroup analysis (Pakzad et al., 2022). This reality limits thorough evaluation of the impact of fairness interventions across skin tone subgroups, unless additional annotation work is done.

GAN-based augmentation approaches are one path towards augmenting underrepresented groups in existing ISIC data - synthetic lesion imagery conditioned based on factors such as skin tone or lesion morphology could be a way to re-balance training sets without collecting over the course of years. However, the phenomena of synthetically-to-realistic domain shifts are still a hazard: synthetic dermatoscopy images also do not allow to reproduce the fine texture cues that exist in genuine lesions, posing a risk of misclassification when applied on real world cases.

From a more general view on the functionality of the dataset, the ISIC series has gradually been extended with not only image data but also segmentation masks for some tasks. The inclusion of binary lesion masks allows for paired image-mask generation methods (e.g., GANs) for associated downstream workflows (e.g. automated lesion outlining prior to classification). Such multimodal contributions increase the potential of SSL methods, e.g. FixMatch using both labelled masks and unlabelled images, that operate efficiently when increasing the training coverage.

Data provenance and annotation quality are some of the common themes in the evolution of ISIC. Large-scale public archives are invariably noisy: non-clinical or mislabelled images embedded in otherwise clean datasets can cause distortion in the distributions of both statistics and learned features (Pakzad et al., 2022). This is important for achieving high levels of diagnostic accuracy while promoting fairness goals discussed earlier in Section 1.1, and such filtering of these anomalies before integration is essential.

A related operational issue, validation realism, is the fact that models optimized over ISIC data may not similarly generalize on other clinical acquisition systems because of imaging device differences or demographic differences. Since SIIM and other collaborations have worked with ISIC to update archives (Das, 2021), it is possible that future versions will attempt to fill some of these long-standing gaps in underrepresentation by coordinating multi-site recruitment to target specific demographics or lesion variants. However, this expansion has led to heterogeneity of imaging protocols, with the necessity for normalization during preprocessing, from resolution standardizing (e.g. rescaling MR/CT analogues in other areas) (Feng et al.) to color balancing for dermoscopic comparability.

ISIC's evolution for cost-effective pipelines needs the integrated use of classical augmentation methods (rotations or flips) with lesion integrity preservation, and of gain-controlled generative approaches (Frid-Adar et al., 2018). Classic affine transformations avoid artifacts as the representation to fix representation bias by keeping the regions-of-interest around lesions; although these operations are limited in their capacity to correct representation bias alone, they are computationally cheap compared to the inference cycles performed by GANs. Synthetic data utility assessments revealed that the use of generated data for augmenting the existing data results in a significant improvement in classification accuracy when assessed by rigorous statistical tests such as paired Wilcoxon Signed Rank assessments (Xing et al., 2023).

Looking to the future, refinement may well depend on augmented diversity along with improved metadata richness (skin tone labels, lesion segmentation), allowing robust fairness aware metric reporting (including ROC-AUC stratified by skin type and calibration curves at fixed specificity thresholds). This is in line with regulatory tendencies toward detailed subgroup transparency while maintaining clinical applicability. The ISIC archive's path, from scatter of first-generation international collection of dermoscopy data through multi-task annotated challenges, hints at further use as a foundation upon which balanced clinically credible AI systems for dermatology can be developed if structural biases are systematically addressed at each evolutionary step.

1.2.2 HAM10000 CHARACTERISTICS

HAM10000 deserves as one of the most widely cited dermoscopic image data set, providing a uniform foundation to validate algorithms and algorithm benchmarking, however, still grow some of the limitations regarding the representation of structure as recognized in Section 1.2.1. It consists of images captured over the decades from several international clinical centers and is integrated into the ISIC archive (Das, 2021).

The quantity of the dataset, as well as the variety of lesion types, including melanoma, melanocytic nevi, basal cell carcinoma, actinic keratoses, benign keratosis-like lesions, dermatofibroma, and vascular lesions, make the dataset an ideal candidate for learning multi-class classification models. Yet fairness aspects are brought up almost right from the beginning of everyone's inspection of the demographic distribution: there is no explicit labeling of patient skin tone or sex in HAM10000, and people have noticed that lighter skin types are over-represented (Pakzad et al., 2022). This imbalance poses a danger that such automated systems optimized on a construct like HAM10000 will underperform for underrepresented people.

From a clinical utility point of view, the image quality provided by HAM10000 is relatively constant because of the dermoscopic acquisition protocols applied in all the participating sites. Lesion images have adequate resolution and color integrity to record fine differences in pigmentation that are important for distinguishing malignant from benign morphologies. However, such uniformity in capture conditions may also represent a limitation in terms of generalization if the target deployment environments may involve greater variability, e.g., capture images with smartphone cameras in community clinics, as opposed to high-grade dermoscopy equipment. This is especially relevant where deployment realism is of paramount importance; where robustness to variation in acquisition settings must be evaluated through cross-dataset validation against collections that represent such variations (Sagers et al., 2023).

Annotation quality within HAM10000 has the advantage of expert labelling pipelines that align histopathological confirmation (biopsy) with consensus assessments by dermatologists. Yet manual annotation requires different levels of time from experts, furthering the economic case for synthetic data augmentation or semi-supervised for easing the burden of annotation (Dhawan & Nijhawan, 2024). Augmentation may be informed by generative models such as GANs or diffusion architectures to up-represent both rare classes of lesions and underrepresented phenotypic situations such as darker skin tones. In particular, it has been suggested that if the new samples can be properly curated in order to avoid the introduction of synthetic artifacts or biologically implausible features, no compromise should be made between the generalization of the classifier and the integrity of the diagnostics (Xing et al., 2023).

The lack of clear subgroup labels in HAM10000 limits fairness auditing. Metrics such as ROC-AUC or PR-AUC cannot be stratified based on the protected attributes without linking the dataset to external metadata sources or retrospective annotation based on inferred variables such as individual typology angle measurements from areas adjacent to lesions on the face. Without such subgroup splits, it is possible for an algorithm to have high macro accuracy, but large differences in sensitivity between light- and dark-skinned patients. Methods based on domain-invariant feature representation learning have shown promise to help alleviate these biases if incorporated into training pipelines incorporating both HAM10000 and labeled data, such as Fitzpatrick17K (Pakzad et al., 2022).

The issue of domain shift from synthetic to real contents becomes a reality if generative augmentation were applied extensively in combination with real dermoscopic images from the HAM10000 dataset. Domain shifts may exist when generated samples have not the microstructures (pigment network or regression structures) which are present in real lesions and which are clinically informative. Pretrained diffusion-based models that learn more complex texture priors than previous GAN models were able to close some of this gap by training on HAM10000-class lesions. Nonetheless, rigid downstream validation is crucial; augmentations that generate high-fidelity visuals may or may not achieve the success in classifier performance without testing them directly on actual benchmark splits.

For semi-supervised learning (SSL) incorporation as a part of processing pipelines based on the HAM10000, algorithms such as FixMatch and MixMatch provide viable means of using unlabeled or synthetically labelled lesion imagery in growing the size of the effective training data without giving commensurate increase in annotation cost. Yet, risks of contamination persist e.g. SSL pseudo-labeling may cause systematic errors to propagate if latent biases persist in initial model states trained mostly on lighter skins heavier datasets like HAM10000 (Sagers et al., 2023). RSMATCH-style adjustments for mitigating such problems have been proposed during the iterations of SSL with strong sample selection criteria based on model confidence distributions.

In addition to raw performance measures, interpretability is also extremely important in determining clinical credibility. With HAM10000 as a training source, the saliency map approaches are able to locate decision-making features within lesion regions, which may also be of use for clinician audit and bias detection, if selected morphological cues can be shown to be effective only among certain subgroups (Thong et al., 2023). Extending this type of interpretability to fairness contexts will be able to show if there is a systematic difference in the breakdown of attributes that cause a lesion between majority- and minority-group lesions when trained on imbalanced inputs.

Cost-effective workflows using HAM10000 might incorporate active learning loops that selectively request expert annotations of the data for cases in which SSL-derived pseudo-label confidence is deemed to be low. Such approaches prevail expert time expenditure while progressively increasing the coverage over lesion subtype under-represented in the dataset baseline. In addition, harmonizing preprocessing protocols, such as color normalization and resolution adjustment, with other datasets also helps address inter-archive distribution shifts for combined training architectures (Sinha et al., 2024).

Overall, HAM10000 continues to be profoundly valuable in virtue of its lesion-type breadth and image consistency but needs strategic supplementation, through generative synthesis that is in line with objectives of fairness, targeted labeling augmentation for subgroup insight, and algorithmic mechanisms for mitigating bias propagation, toward fulfilling the expectations of modern clinical deployment across diverse populations at acceptable cost levels. Integration with larger collections of data that have explicit annotations of protected attributes promotes both technical fairness metrics reporting and substantive compliance with regulatory advice under HIPAA or GDPR privacy frameworks (Raju et al., 2024).

1.3 REPRESENTATION GAPS AND BIAS

1.3.1 UNDER-REPRESENTATION ACROSS FITZPATRICK SKIN TYPES

A general problem in dermatology datasets is the imbalance in distribution of Fitzpatrick skin type (FST) spectrum, which leads to unequal diagnostic performance that directly correlates with fairness issues that we discussed in Section 1.2.2.

India: Important Image, Skin, and Color Imaging Data Sets - Judith Towards Findings: The vast dyads such as HAM10000 and Isic Skin Image Cloud (ISIC) often map the prejudices light skinned photococ_field, which are often mirrored in white clinical image capture facilities that over-represent people of European descent.

Quantitative surveys based on a single typology angle measurement or direct FST annotation support the fact again and again that images corresponding to Fitzpatrick I-III clearly predominate and types IV-VI are rare. This difference impacts the visibility of lesion-level morphological features (Pakzad et al., 2022), as well as the overall training distribution to the models.

This results in the following situation: A model trained largely on lighter skin lesion specimens will be expected to learn decision boundaries that will not generalize under the darker skin scenarios, where the contrast between color differences are less distinct and the contrast ratios varied. We also find that anisotropic representations of learned features lead to decreased sensitivity values and increased false-negative rate when there is lower representation of specific FST categories.

The clinical implications of this mismatch are not theoretical; the lower sensitivity in higher number Fitzpatrick types undoubtedly increases the risk of missed detection of melanoma in the darker-skinned population. The economic cost is derived not only from the worsening of the disease but also from possible loss of confidence between patients and institutions using AI-based triage tools.

The literature emphasizes that even after post hoc debiasing strategies implemented, even datasets do exhibit performance losses, as evidenced by face recognition tasks and diagnostics classifiers (and numerous others) (Wang et al., 2022). If attempts are made, however, to reduce lesion recognizability via rebalancing counts, there may be a suggestion that the persistent difficulty arises from inherent optical phenomena such as diminished thanks to lower luminance contrast between lesion and background skin.

Under-representation is also quantified differently depending upon the chosen skin tone quantification methods. Most skin color fairness analyses have utilized a single tone axis (L) usually describing the lightness/darkness without taking into account changes in hue due to aging or photodamage. However, apparent skin tone is a multidimensional characteristic in terms of hue, saturation, and brightness; using only tone is unable to describe the yellowness increase in older Asian dermatotypes or the reddish changes in aging skin of the Caucasians (Thong et al., 2023).

Nevertheless, if imbalance metrics reflect such fine discriminations and put these in coarse bins that rely only on Fitzgerald's original schema, correction controls will fail to account for clinically relevant heterogeneity. Multidimensional descriptors of apparent skin color have the potential to both enhance bias auditing by linking subtly colored chromatic signatures to model error distributions rather than by compressing all color variation down to gradations of lightness.

The use of overall artificial data augmentation shows potential to be effective in closing representation gaps by purposely creating lesions on darker-skin backgrounds. GAN-based implementations offer the possibility of conditioning on FST labels or inferred tonal attributes to generate samples explicitly tailored for less represented 'odd' types (Georges-Filteau & Cirillo, 2020), whereas diffusion models provide richer interpenetrative texture, possibly addressing issues of artefact danger noted with past GAN results (Sagers et al., 2023).

For instance, when injected into small real datasets, the performance of synthetic imagery injection in the context of wound segmentation research was shown to be equivalent to much larger real-only datasets (Sinha et al., 2024). Application of our findings to future skin cancer classification does come with strong limitations: synthetic-to-real domain shift is also a severely feared threat unless these generative models successfully reproduced also the microstructures (like pigment networks), typical of real lesions. Because of this, the marginal benefit from increased representation may be cancelling out cluster poor classifiers precision during production.

Also, how domain-invariant representation learning methods can play a role in partially bridging these gaps is also considered. By also enforcing the invariance of protected attribute like FST score during the feature extraction stages with learned transformation between skin types, the classifier can have more consistent grouping geometry even without literal parities of dataset (Pakzad et al., 2022). While this may be a solid way to improve the ability to generalize when minority-skin samples are rare, it is complex as it may be too abstract away from actual tonality information which may harm clinical interpretability or blunt relevant discriminative signals carried by pigment changes.

Semi-supervised learning (SSL) opens the path of balancing the effective sample mass for the darker skin categories without equal manual labelling cost. Autoencoders for data augmentation: new deep learning methods like FixMatch or MixMatch are able to take advantage of unlabeled lesion images collected in the clinics of high-FST populations to cost-effectively increase coverage. Nevertheless, if pseudo-label propagation passes biases

from the original (trained mostly on light-toned) models, incorrect labels can become more and more contaminated; remaining ST-based pseudo-label filtering strategies like RSmatch are used as mitigation against accident contamination during the SSL training iterations.

SwingReverse outcomes are further improved by having active learning loops which narrows down the set of minority types of cases to be labelled by experts based on model uncertainty trends, focusing labeling spend in areas of representation where labeling is most likely to positively impact fairness metrics.

Stratified validation is also needed across true FST bins for datasets that are balanced via synthetic augmentation for deployment realism aspects. The use of metrics such as the area under the receiver operating character (ROC-AUC) curve disaggregated by Fitzpatrick type and sensitivity at a proposed cut-off threshold of 90% specificity can translate into assessing whether an augmented diversity translates into equitable predictions or just an improvement in the global mean that accounts for prevalent subgroup disparities (Sagers et al., 2023)

Calibration metrics also provide data on whether probability outputs have stable true outcome probability content in varying tonal backgrounds, which is an important property since the score should aid clinician prioritization for biopsy referral.

From a regulatory perspective, the need to report on performance by subgroup-level characteristics in training corpora is in sync with standards such as GDPR and HIPAA that are emerging to require such transparency. In addition to source provenance, the inclusion of protected attribute annotations make it easier for clinical safety audits to identify the remaining domain of bias before implementing models (Thong et al., 2023).

One example of this is the infusion of multidimensional apparent color scales into default pre-processing rendering a consistent input representation to models across capture conditions, illumination correction, and device standardization (especially where acquisition hardware etc varies between contributing sites). This will require intersecting layers of more data; quality/pigmentation skin-tone controlled synthetic generation; ERS skin-tone invariant feature extraction; SSL and active learning in ethically generated data loops; stratified ROC-AUC and calibration reporting; and patient safety in product design and post-deployment monitoring.

Due to the physical constraints, it is impossible to provide a valid, reliable diagnostic without introducing a risk of patient mistrust, which drives convergence of the technical and operational countermeasures that are needed to bridge the Fitzpatrick gap (Wang et al., 2022).

2. SYNTHETIC DERMATOLOGY DATA GENERATION AND LABELING

2.1 GENERATIVE MODELING APPROACHES

2.1.1 GAN-BASED SYNTHESIS

Generative adversarial networks (GANs) have been extensively researched for synthetic dermatology image generation, both to make a dataset balanced (e.g., higher Fitzpatrick skin types are underrepresented and GANs help to synthesize more of them) and to decrease human annotation costs. In its classic form, the GANs are a combination of a generator and discriminator in an adversarial training setting, where the generator looks for the real output generation and the discriminator separates the real and fake samples.

For dermatology datasets such as ISIC or HAM10000, conditional GAN frameworks are applicable and they can be modified to explicitly control for attributes such as skin tone, lesion type, or acquisition modality. These conditions are directly compatible with policy interventions to increase fairness in the skin matching sub-groups covered in previous analyses. Conditioning on metadata-inferred or tone-based metrics allows to condition on very specific minority-class, and underrepresented phenotypes and generate content with unprecedented efficiency in the acquisition of new real images at prohibitive cost.

Refined versions of the variation (THF) generators are encoder-decoder CNN structures that sample points in latent space around real samples and induce "neighborhoods" of images that resemble the originals but have sufficient variance to show clinically relevant outcomes (Georges-Filteau & Cirillo, 2020). This kind of synthesis is inherently mode collapse-resistant and the transitional distributions produced are richer than would be possible with the unconditional architectures, which can enhance utility for models trained on output. Because of the ability of these synthetic products to be used within semi-supervised learning (SSL) pipelines like FixMatch or MixMatch, the results of these pipelines can be used to improve generalization in the case when labeled data are difficult to obtain, particularly for rare lesion types or darker skin tones.

As pseudo-labels result from biased seed models, however, caution must be exercised in establishing the pseudo-label propagation connectivity; if not cleaned through confidence-based criterion similar to RSMATCH, it may cause the pseudo-labels from different groups to strengthen existing discrepancies. From an architecture perspective, medical imaging studies have also benefitted from architecture changes such as SPADE layers which incorporate spatially adaptive normalization into the generative process (Pinaya, 2023).

By adjusting normalization parameters based on semantic mask, e.g., lesion segmentation maps, SPADE-enhanced GANs retain fine-grained facial details within the disease areas but allow for the surrounding skin tone to vary in a realistic manner. This is valuable not only for enhancing apparent realism, but also for drawing focus points when performing fairness audit, i.e. determining whether model saliency is focusing in the right places for lesions and not background artifacts. Integrating the patch-based discriminators in these pipelines further increases the local texture fidelity by assessing the smaller receptive fields instead of the whole-image realism only, which is important for medically relevant microstructures such as pigment network or regression areas.

Fairness objectives overlap with best practices for clinical accuracy during the purposes of synthesis across multilayer tonal backgrounds. Though training corpora can be synthetically balanced on Fitzpatrick-type identifiers or continuous individual typology angle values (Georges-Filteau & Cirillo, 2020), this is predicated on the ability of the generative model to accurately mimic chromatic shifts that are specific to darker skin, which classical GANs have difficulties with. Diffusion-based textural priors have shown to exceed the performance of earlier GAN formulations at replicating these patterns (Sagers et al., 2023), so hybrid methods may help to reduce risks of generation artifacts at high FST while maintaining reasonable compute costs with reuse of pre-trained GAN components which are structured on dermatological images.

Using GAN outputs solely for clinical applications still introduces the potential for domain shift between synthetic and real inputs (also called the "synthetic to real 'paradigm shift'"), which is still a fundamental risk for training classifiers that will be eventually deployed in clinical workflows. Empirical studies show that, in spite of some synthesised images obtaining a high visual fidelity, they can still fail utility tests since they lack subtle but yet diagnostically relevant features existing in real lesions (Xing et al., 2023). Regarding the application of such models, assessments should include not only global measures such as ROC-AUC or PR-AUC in blend datasets but subunit performance in a subgroup-based evaluation (ROC-AUC per Fitzpatrick type at 90% specificity) as well as the calibration curves stratified by skin pigmentation group (Sagers, et al 2023). Even if desired accuracy results rise worldwide after augmentation, failure to validate the subgroup may erode the clinical trust.

Another aspect with influence on the realism of deployment can be found in computational efficiency and related cost profiles at synthesis stages. Training heavy GANs on large image datasets requires significant GPU memory capacity, as well as long training times (Xing et al., 2023). Transfer learning models (TLLMs) pretrained on extensive domains such as GANs with general visual structure knowledge are often used to decrease costs by only fine tuning the model to specific parameters required for domain adaptation to dermatology images.

For instance, from such pre-trained backbones, using biologically based design we would be able to extract synthetic unlabeled data to permit running standardization pipelines via an end-to-end closed loop solution, removing the privacy risks associated with using patient data directly while reducing annotation costs. Ability to synthesize patient-specific data: Privacy protection is another driving force for the synthetic data integration: GAN-based synthetic generation of anonymization-fit and clinically relevant samples overcomes potential institutional review board's restrictions associated with anonymized images (Yamaguchi 2023). Again, fidelity is of critical importance; if privacy-safe samples do not include sufficient diagnostic information, utility objectives would be compromised even if they do meet anonymity constraints.

This is especially sensitive under regulatory frameworks such as GDPR and HIPAA, whose requirements even pertain to patient privacy protection, despite generative augmentation being applied - this is for extending medically relevant morphology to patient demographic ranges without compromising them via inversion attacks and model-leakage experiments. As GAN-synthesized augmentation is incorporated into the development workflows of dermatological AI, it is important for it also to enjoy MLOps aligned controls in its implement, including modules of artifact detection during generation run-times. Quality screening based on classifier-based realism measures with rejection sampling at the generation stage may ensure that outputs containing obvious anatomical photomontages and generative hallucinations are discarded before being fed into downstream training sets.

Thus, the control is particularly critical when trying to learn a target population that is a minority representation where the scarcity of samples increases the relative impact of each poor synthetic data point in biasing learned distribution boundaries toward a spurious correlation. Finally, active learning loops combined with synthetic augmentation strategies proved more cost-effective: generated candidates with the highest likelihood of improving fairness-adjusted utility could be shortlisted for human annotation before being added to canonical training sets (thereby effectively prioritizing physician annotation efforts).

Doing this connects generation effort deflection (control) closely with downstream performance objectives while maintaining subgroup equity originals set in validation planning phases for regulated sources (Thong et al., 2023). Unless these mechanisms are coupled, the synthetic conditioning, stringent stratified evaluation after integration, SSL-aware debiasing filters, privacy consistency with the law, the benefits of GAN synthesis risk would be outweighed by domain drift and unquantifiable bias amplification between protected skin-tone classes to be represented.

2.1.2 DIFFUSION-BASED MODEL FOR IMAGE GENERATION

Especially when high fidelity and texture realism are important for clinical downstream applications, diffusion-based generative models have emerged as a competitive alternative to GAN frameworks and generate synthetic images in dermatology. Unlike adversarial training, diffusion methods diffusively restore samples from a Gaussian distribution to target images and are able to model fine-scale structures like pigment networks and regression regions, that tend to be difficult for GANs to reproduce correctly (Tariq et al., 2023).

This characteristic is directly relevant to the problem of representation bias presented by datasets such as HAM10000 and ISIC, which have very low representation of higher Fitzpatrick skin types, thus hindering model performance across populations. By conditioning on tonal descriptors such as explicit Fitzpatrick or continuous individual typology angle, diffusion pipelines are then able to synthesize lesion presentation over a broad chromatic spectrum, without the same severity of mode collapse as conditional GANs (Sagers et al., 2022).

A useful benefit of diffusion techniques in this regard is their ability to support text-to-image coupling or semantic conditioning using prompt engineering. This enables a combination of dermatologically valid restrictions, type of lesion, anatomical region, skin colour pattern and skin surrounding and preserves clinically valid morphology.

In a case study aiming at fairness goals, a generation run meant to boost the representation of Fitzpatrick V-VI melanoma can encode both pigment depth and morphological characteristics into the synthetic image pool. The

voice-regularization applied in the iterative denoising process lowers the occurrence of high-frequency artifacts such as checkerboard or color banding previously reported in previous GAN approaches on darker skin tones.

Moreover, empirical tests have shown that these spatial generalization improves when these models are synthesized with lesions on different backgrounds, this is due to the fact that we might be deploying them on a different acquisition machines and different illumination conditions (Tariq et al., 2023).

When applied into semi-supervised learning pipelines, diffusion outputs can be used both as augmentations of unlabeled data with pseudo-labels in methods like FixMatch or MixMatch as well as real-label augmentations of rarely identified FST-classes after manual dermatologists' annotation. The main limitation here is contamination risk; if pseudo-label propagation is highly dependent on a base model trained on a dataset with class imbalance (e.g. dominated by only Fitzpatrick I-III), the artificially generated inputs, even if very realistic, might transfer label bias into subsequent rounds of training.

One mitigation approach takes a page out of the RSMATCH playbook and considers prediction confidence across multiple augmented views before accepting pseudo-labels for SSL update (Sagers et al., 2022).

From a cost performance perspective, the diffusion models are generally computationally expensive compared to a similarly scaled GANs because of the multi-step denoising procedure. Technical improvements, including using smaller number of timesteps, diffusion backbones pretrained on medical datasets (Tariq et al., 2023), and additional model adjustments result in smaller computational footprints while preserving image quality.

For expense-constrained environments where cost is a more critical factor, hybrid schemes in which a diffusion model is used to synthesize a small high-fidelity reference set and interpolative lightweight architectures are used to synthesize variations from it can potentially provide an acceptable synthesis cost/augmentation benefit.

The clinical feasibility of such approaches necessitates the verification that downstream classifiers retain performance equivalence with complete diffusion augmented training regimes with respect to subgroup specific ROC-AUC and sensitivity-at-90% specificity metrics (Sagers et al., 2023).

The key point is that synthetic sets need to be stratified during validation stages (FST distributions) and ROC-AUC scores per-skin-type, and calibration curves, want to be monitored against each other so that confirmation of the absence of bias on the global level does not hide the fact of any bias on the skin-level. Such attendant inter-tone calibration discrepancies can result in under- or over-triage in clinical practice where such probabilities are used to make decisions regarding biopsy referral (Thong et al., 2023).

Moreover, as lesions generated through diffusion have lower rates of artifact than most GAN outputs, anomaly detectors would yield fewer false positives during automated QC; nevertheless, hallucination risks remain, especially when the conditioning set fails to include rare but clinically relevant lesion subtypes like amelanotic melanomas. This is where automated histopathological plausibility scoring systems, or multi-expert dermatology review panels to, become important gating steps, before they can be part of production training sets.

Data privacy issues overlap naturally with the implementation of synthetic dermatological content under GDPR and HIPAA regulations. DAG gives rise to images that are purely based on the statistical priors estimated from the training data as aggregated from many patients, which provide a compelling direct surrogate for real patient images while minimising direct re-identification risk (Sagers et al., 2022).

Still, safety tests, which are required by regulators, should be performed to ensure that inversion attacks cannot recover the source images in an identifiable way from weights learned from the models, which is a concern in generative architectures, irrespective of the flavor (also known) in the literature. Other operational safeguards include implementing watermarking as part of all synthetic outputs for traceability in MLOps pipelines that encompass long-term maintenance of clinical AI systems.

The existence of domain shift between synthetic and real is still central to the problem, even though various iterative denoising works have been developed. While the quality in local texture reproduction for the diffusion-based images is excellent, the images may reflect shade variations or incorrect shadowing in a certain tonal range if the conditioning corpus was not sufficiently diverse in illumination. Such variations have the potential to mildly compromise reliability of classifiers when scaling from the laboratory to field applications with various heterogeneous imaging devices (i.e., dermatoscopes through to smartphone macro lenses).

As seen in other verticals within medical imaging that utilize domain adaptation to overcome modality differences between MRI/CT (Choudhary et al., 2020), the same transfer learning adaptations may need to be made in alignment with stylistic differences between diffusion images and photographed lesions before consumption by the model. However, the use of domain adaptation steps after synthesis, or of co-training discriminators on real and generated distributions of data, opens new possibilities for reducing mismatch effects without losing potentially useful synthetic diversity.

Overall efficiency development from such generation strategies would be in sync with a full workflow when mapped to cyclical learning tied directly to fairness measures from audits during periodic validation. Giving primacy to generation requests for particular lesion-type/skin-tone combinations that drive measurable subgroup performance deficits, and decommissioning generation targets after validated parity thresholds are met, makes it possible to bound compute cost while moving toward equity-aware deployment maturity faster than unguided augmentation policies (Thong et al., 2023).

In combination with the iteration by diffusion modeling and a targeted selection coherence derived by clinical error analysis, this concept stands out as a possible successor to the conventional GAN-based augmentation for addressing both the fairness gap and realism limitations on AI reliability in dermatology across demographics.

2.2 DIVERSITY: SYNTHESIZING STATISTICS FOR DIVERSITY

2.2.1 STIMULATING SKIN-TONE VARIATION BY WAY OF PROMPT ENGINEERING.

Controllable Generation with Transformers - Using Ising Boosted Generative Adversarial Networks (IBGANs) for Synthetic Dermatology - via specificity prompts to guide images towards a specific skin-tone distribution will help to bridge the gap that is thought exists for higher Fitzpatrick types in HAM10000, ISIC archives, Fitzpatrick17K, etc. Together with the modern text-to-image diffusion architectures or conditional GANs, prompt engineering is a fairness intervention as well as a data augmentation technique.

The aim is to incorporate tonal variability without sacrificing lesion morphology of clinical diagnostic relevance, particularly in minority sub-populations where training is limited which has association with loss of sensitivity.

By placing manifest structures such as "melanoma on Fitzpatrick type V skin" or "basal cell carcinoma on deeply pigmented epidermis under bright clinic illumination" into the conditioning channel, it is possible to systematically fill previously low-density regions of the tone-hue distribution space.

One of the important aspects of building such prompts is the representation of skin color. Coarse categorization through semantic labels such as Fitzpatrick Type misses on hue nuances (e.g. reddish versus yellowish undertones) that have been proven to affect algorithmic performance independently from lightness. More advanced types replace a static descriptor by the continuous values estimated using expressions such as an individual typology angle L^* threshold or a hue angle h^* , in which, for example, $L^* > 60$ means lighter colors and $h^* > 55^\circ$ means yellow, and uses that information to guide the generator (Thong et al., 2023).

The direct incorporation of both dimensions into the prompt templates leads to more accurate approximation of the multidimensional distribution of real-world skin chromaticity by model output. Without subtle control of the hue, it is likely that the resulting synthetic sets will look superficially diverse when only lightness is considered, but will not represent the clinically relevant shade variation across ethnicities and with age.

From an operational standpoint, timely engineering is highly compatible with the cost-efficiency imperative. It allows for finer-grained targeting of demographic gaps from synthetic batches, drastically reducing the amount of collection campaigns that need to be built in the real world. Conditional GAN architectures can tie text discourse cues into latent variable space conditioning; diffusion can also incorporate them into GAN architectures by cross-attending in denoising steps (Tariq et al., 2023).

This flexibility enables iterative generation approaches where prompts are re-tuned during the middle of the cycle based on interim fairness assessments (e.g. ROC-AUC per FST bin / calibration curves per tonal sub-group) and data generation budgets can be redistributed dynamically towards the worst-performing classes. However, tone control over direct text also brings with it risks that must be tracked inside MLOps pipelines. Sometimes, even when asked explicitly to do so, these models may have a difficult time accurately rendering darker tones, resulting in images with artificial color banding or decreased lesion visibility. This becomes especially troublesome when

rare lesion classes are synthesized in high FST environments, as non-obvious microstructures (e.g. pigment networks, regression areas), are prone to being washed out in the transition from the semantic description to the image output (Sagers et al., 2022).

A combination of post-generation screening via automated detection for artifacts and dermatologist review (where possible) can be used to screen out low utility images from potentially contaminating the loop of SSL training like FixMatch or MixMatch. Without precision calibration, pseudo-label propagation from biased seed models may well instill spurious correlations into underrepresented groups, compromising fairness goals despite an increase in apparent diversity.

Integration of metadata mined from existing corpora is immediately useful in the project of construction itself. Based on studies that analyzed large datasets of facial images, it has been shown that there is bias towards light-red skins and lack of dark-yellow combinations (Thong et al., 2023), and dermatology datasets probably share these imbalances.

By creating prompt libraries that specifically aimed for under sampled junctions (e.g. Fitzpatrick VI coupled with high h^* yellows), synthetic generation can force saturation of specimens that may have been rarely found in archival collections. Adding more contextual stimuli to prompts, lighting conditions, anatomical site descriptions, etc. not only challenges the generator with respect to co-variations, it brings downstream classifiers to more deployment-relevant acquisition environments.

Another pragmatic layer is that of prompt-adaptive sampling: after the first batch generation with the balanced template set, the real-time bias audit shows what the particularly effective subset of prompts should be upweighted with in the subsequent cycle AFTER observing the subgroup performance changes. For instance, if the threshold of ROC-AUC for Fitzpatrick IV-V melanomas was about to reach a plateau, but for type VI basal cell carcinomas it was still a daunting lower risk threshold, the emphasis would quickly shift in that direction. This feedback mechanism curbs redundant synthesis if parity has already been achieved and redirects resources to lagging strata, a real-world tradeoff path between cost effectiveness and equity goals.

Another issue of privacy is that prompts may be used to detail demographic contexts without necessarily naming specific individuals or creating full copy cases from sensitive data sets (Akroun et al., 2023). This should be of value for GDPR/HIPAA compliance, as images generated through the method will be by statistical prior aggregation, rather than single-source memorization, which is a crucial difference when using AI systems across institutional boundaries.

Nevertheless, inversion risk audits are still recommended, as even synthetically generated high-FST lesions have to be checked against re-identification attacks (to make sure no unforeseen leakage from the pre-training data has happened).

Clinical plausibility within the generated diversity should be achieved also by directly integrating domain-specific constraints in the prompt syntax itself. While skin tones can be described as shades of blue, green, brown & grey, references to histopathological correlates (i.e. "irregular pigment network", "blue-whitish veil pattern") ground generation in clinically relevant attributes instead of superficial skins cape variation. Consequently, the diffusion process tends to minimize overrepresentation of background color at the expense of diagnostic lesion properties, which is the most frequent point of failure in diffusion where over-representation of attributes not directly corresponding to a lesion is used without proper context.

A closed-loop feedback loop between specific conditional descriptors and the dynamic synthesis planning is then shown to allow for effective deployment of skin-tone variability prompt engineering. By enabling the combination of the textual specificity in hue-tone-space agnostic controls and powerful quality gating stages and aligning them to fairness metrics recorded during development, it is possible to extend the representation in a meaningful way while minimizing cost overruns and bias amplification risks (Wang et al., 2022). The ultimate goal is not only the photorealistic diversity but the diversity that is credible clinically and preserves the diagnostic integrity among the demographic categories at the point of care.

2.3 LABELING TECHNIQUES AND COST SAVING

2.3.1 WEAK SUPERVISION METHODS

Weak supervision is a practical approach for alleviating annotation burden in pipeline dermatology while maintaining the clinical integrity required for the skin cancer detection task. In contrast to fully-supervised methods, which rely on exhaustive expert annotations, these methods make deliberate use of noisy, incomplete or indirect annotations, making up for their reduced fidelity with algorithmic methods that can mine informative learning signals from them.

Weak supervision becomes particularly appealing when carefully considering the fairness aspects and deployment realities, especially in the face of the limitations of many data sets such as ISIC, HAM10000, and Fitzpatrick17k, which suffer from a representation gap and the cost of labeling.

In one class of weak supervision techniques, we can address this by generating pseudo-labels from models trained using a small number of high-quality expert-labeled examples (Mu et al., 2020). In the case, the training is a model that predicts labels for unlabeled images, which are used as training targets for further iterations.

While pseudo-labels enable swift data augmentation at no additional labeling cost, they do present the risk of contamination, particularly when source models encode the bias related to over-representation of light skin tones, or specific lesion classes. If unattended, error propagation may entrench subgroup disparities.

One of the tricks is confidence-based filtering, as used in RSMATCH-type methods where only predictions above adaptive thresholds are kept (Yamaguchi, 2023). This adaptivity decreases mislabeled minority-skin-type accumulation in the validation data, which otherwise degrades subgroup sensitivity.

Other variants incorporate consistency regularization in the pseudo-labeling process (Mu et al., 2020), which enforces consistency between a model's predictions under two or more perturbations of the image (e.g., applying some geometric transformation, jittering colors, etc.). This further limits the decision boundaries to more stable characteristics (like lesion morphology) and not transient cues like illumination or background artifacts. Such methods also overlap with fairness objectives promoting invariance to features correlated with protected attributes such as skin tone or sex, although tuning must be used to make sure clinical detail is not smoothed out in the name of invariance.

Weak supervision can also come from auxiliary label sources like pathology reports, dermoscopy screening logs or even patient self-reports from mobile health apps. While these annotations usually are not as precise as they are when dermatologist-assigned image labels, they can be matched with images using natural language processing pipelines and entity resolution algorithms. Because they are less expensive per sample than are formal biopsy-confirmed labels, they allow high throughput injection of large numbers of cases that are underrepresented in the patient subgroup from which these samples were obtained into training corpora.

However, there is a need to carefully calibrate matching textual descriptors against visual features; potentially ambiguous descriptors should be carefully pre-filtered to maintain consistency in lesion location using downstream checkers or automated scoring of image-text embeddings similarity before being included in the training set.

Heuristic labelling is another example of a weak supervision paradigm. In the domain of dermatology, this process could be a rule-based classifier that uses color histograms or border irregularity measures which are validated by the experts to first screen large image databases (Baugh et al., 2023). Although such heuristics are too brittle to provide a final diagnosis as they break down in atypical morphologies, they can provide rough labels that are used to train the SSL frameworks such as FixMatch and MixMatch for refinement. By up-fronting coarse screening towards demographics with limited representation currently in archives (e.g. Fitzpatrick V-VI), collection efficiency increases without proportional increases in specialist effort.

The generative augmentation pipelines described above can also be used as additional input channels for weak supervision loops in case synthetic images can be linked with "noisy" labels that result from conditional generation parameters instead of human inspection (Yamaguchi, 2023). For example, in a GAN conditioned on melanoma for Fitzpatrick skin-tone VI synthesis, the output samples contain by-construction the target class, even though instances will not be clinically plausible due to the imperfection of GAN generation. These synthetics, which are noisy labelled, are suited for SSL training after purifying them with discrimination modules, which would reject

outputs that are too implausible (Dhawan & Nijhawan, 2024). Synthetic datasets focusing on minority groups can be implemented at increasing scales using such mechanisms, but fairness-aware performance metrics (ROC-AUC stratified by Fitzpatrick type) will need to be tracked at all times to ensure that representation improvements do induce classifier generalization gain.

Another interesting direction is multi-instance learning (MIL), which classifies bags of images rather than individual images. In the case of a dermatology clinic this might involve applying a "positive" class label to all images in a patient record if any biopsy confirms malignancy (Choudhary et al., 2020). However, the MIL algorithms are set to recognize discriminative instances in the bags and mitigate noise during training, so the only potential weaknesses arise because of benign lesions within the bags are erroneously labeled positive. Stratification of patients in terms of skin tone and sex might be used to distribute MIL over patient records to maximize the coverage of subgroups with a minimal computational requirements.

Computational efficiency also comes into play: weakly supervised methods are often able to be done without pixel-perfect segmentation requirements (Abir et al., 2024), thus saving both GPU hours as well as dermatologist annotation time compared to dense mask creation. This is in accordance with the deployment realistic goal, where models constructed from lower granularity supervisory signals are expected to generalize more effectively to acquisition environments that differ in the hyper cameras and conditions because of a non-overfitting problem.

These strategies should have multi-metric validation tests and not simply published mean improvement on pooled data to demonstrate clinical validity. ROC-AUC and sensitivity-at-90% specificity scores, disaggregated by Fitzpatrick type and sex subgroups (Pakzad et al., 2022), and calibration analysis to determine if predicted probabilities have the same meaning across all subgroups of the demographics. Without such disaggregation, weaknesses in supervision that target particular underrepresented groups may be concealed and overall improvement in supervision may lead to an overstated perception of the impact of weak supervision.

Both real-life and model synthetic data may contain weak domain shift that suggest inclusion of hybrid domain adaptation steps in SSL workflow (Choudhary et al., 2020). An adversary-based feature-space adaptation or transformation approximation can help align distributions between noisily labelled data from one source domain (e.g. smartphone-acquired community clinic images) against expert-labelled dermoscopic datasets such as HAM10000 that can serve as anchor references for lesion morphology standards.

In MLOps practice, weak supervision modules should be placed in operational active learning loops that, based on the model uncertainty on incoming unlabeled pools, can promote prioritization of what samples get promoted from weak to strong labels through dermatologist review, for maximum marginal benefit per dollar spent. This cascades through multiple cycles, both balancing representation and achieving robust model performance without exceeding costs determined by sustainability thresholds of deployments in privacy compliant environments such as GDPR/HIPAA (Yamaguchi, 2023).

Fairness-aware performance monitoring, in tandem with adaptive sampling criteria, make sure that improvements based on cheap but noisy label needle do not hurt trust or make the gap worse when systems are moved from the laboratory marking to production sets of dermatology workflows.

2.3.2 IMAGE ANNOTATION FOR DERMATOLOGY USING ACTIVE LEARNING

Active learning provides a focused method of minimizing annotation costs in dermatology AI workflows while at the same time working to achieve fairness goals and clinical readiness. The key idea is to sequentially sample the most informative samples from large unlabeled pools, i.e. samples that will improve the model (by an expert annotation) the most, and prioritize them for annotation. Besides the potential time-saving physician label in the real world, such automation will concentrate labeling efforts on underrepresented groups such as Fitzpatrick IV-VI or female patients when there is a demographic imbalance in datasets such as ISIC archives, HAM10000, BCN20000, or Fitzpatrick17k (Pakzad et al., 2022). By intelligently choosing the samples that need expert review, active learning can speed convergence toward fair behavior across the sub-groups without the need to cover the unlabeled corpus in its entirety.

Uncertainty quantification plays the role of a key component to the selection strategies for active learning. Common metrics include entropy of class probability distributions, margin sampling (difference between top two predicted probability) or variance of ensemble class predictions. In the context of dermatological image classification, fairness-compatible querying by entropy is compatible with per-subgroup stratification, and can identify cases with high uncertainty in the context of underrepresented tones or lesion types to be surface (Wang

et al., 2022). This gives a counterweight to the possibility otherwise achievable through purely global uncertainty sampling that overrepresented categories could be targeted by the sampling more often due to the higher complexity of their decision boundaries caused by the higher volumes of data.

A variant is the cluster-based active learning approach, which uses clustering in feature space, like k-means clustering, to encourage that the samples that are learned from are from the unexplored parts in the distribution. This approach is especially applicable to unsupervised balancing across Fitzpatrick types as RGB/huespace clustering can help concentrate on tonal regions that are low in density in the present labeled data (Thong et al., 2023).

Cost-efficiency can be improved by combining it with semi-supervised learning: pseudo-labels generated by SSL can be tracked for confidence decay patterns, and where pseudo-labels seem unstable, active learning queries can be sought - pseudo-labels are considered the 'feature', and the decay pattern is the 'label' (Sagers et al., 2023). This hybrid approach ensures that costly dermatologist reviews are applied only to cases of high disagreement between SSL predictions and model priors and belonging to underrepresented demographic subgroups, extending fairness to all patients and concurrently improving global accuracy per labelled example.

However, if SSL pseudo-labels feeding into sampling are contaminated by systematic bias, there is a risk of reinforcement loop; using RSMATCH-style filters, pseudo-label candidates below dynamic confidence thresholds are removed so as to reduce erroneous reinforcement loop (Yamaguchi, 2023).

From an MLOps perspective, active learning modules can be integrated with demographic metadata monitoring so that query rates per subgroup can be verified against fairness benchmarks. For instance, when the calibration sensitivity for Fitzpatrick type VI melanoma is lower than for type II after several cycles of a deep Q-learning model capturing improvement in ROC-AUC (Sagers et al., 2023), the query policy dynamically up samples instances from minority-tone clusters until parity thresholds drift toward acceptable levels. In addition, an institutional dashboard with real-time feedback associated with regulatory compliance factors mandated by GDPR/HIPAA for subgroup-level reporting can be integrated (Thong et al., 2023).

Clinical validity places limitations on selected heuristics beyond the probabilistic ones. Generative augmentation pipelines which are fed into unlabeled pools without quality control, overfit on atypical artifacts like synthetic halo effects or compression artifact, resulting in their overrepresentation (Yamaguchi, 2023). Automated artifact detection modules can rank candidate images prior to inclusion in active query shortlists (poor examples identified here should be demoted, no matter the uncertainty value unless artifact classification itself is an ancillary training goal (e.g. for robustness in production). Additionally, GANs or diffusion mechanisms discussed previously in Section 2.3.1 for generating artificial lesions within synthetic data exist, but their use is limited because, without being equipped with the ability to detect hallucination within the stream, they will waste a dermatologist's time.

Checking the fairness of assessment tools and the realism of the deployed system must be intertwined with the process of performance evaluation for validation in the active learning lifecycle. The characterization of ROC-AUC/ PR-AUCs in stratified by Fitzpatrick/sex suggests informative measures of whether rounds of targeted annotation are bringing fairness or accidentally pushing benefits into dominant strata (Pakzad et al., 2022). Calibration curve analyses determine whether predictive certainty is consistently aligned with true probabilities across demographic groups, a feature which is critical when AI outputs progressively determine urgency for biopsy referral in the clinic. Examined through the lens of clinically acceptable false-positive rates, subset parity allows sensitivity of 90% or greater while maintaining 90% specificity, making effortful parity not overly burdensome in the clinical workflow.

Cost modelling of the implementation of active learning is contrasted with the overall annotation campaigns: the review hours required of the physician are roughly proportional to the size of the queried set and not to the full volume of the dataset. From the perspective of cost-effective clustering, this can be an order of magnitude less required expert time when cluster-diversity conditioner and SSL-assisted preselection cells can be used to achieve the same improvement on equity-adjusted accuracy as exhaustive labeling would. This proportionality is significant when scaling up to archives of hundreds of thousands of dermatology images aggregated across institutions; privacy regulations often only allow sharing of raw images in house but sample targeting based on metadata when sharing outside the organization (Abir et al., 2024). Model maintenance costs are a component of the costs of solution implementation and once the model has been deployed there is a need to maintain the model and maintain investment must be justified based on cost-benefit analysis using labeled-per-performance-gain ratios.

Even here, real-world performance is affected by domain shift considerations. Unadapted ingestion may lead to a model's generalization to be degraded (Choudhary et al., 2020), since active learner queries in the clinic and acquired via non-dermoscopic means (such as smartphone photos) may reside in a feature space which is distinct from the one pre-trained on HAM10000 dermoscopy. One such mitigation is the adoption of domain-adaptive preprocessing, color constancy corrections, resolution normalization, to candidate query images prior to sharper application in model's uncertainty evaluation pipelines. Alternatively, adversarial domain alignment techniques can normalize feature representations before the uncertainty is calculated so modalities of capture do not skew selection priorities away, based on the differing importance of different cases to minority subgroups.

Finally, the addition of active learning with generative augmentation is efficient: if diversity-centered prompt engineering produces realistic synthetic samples reflecting gaps (as described above), they may be used as "surrogates" to fill feature neighborhoods around minority real-world cases, before real querying sessions (Tariq et al., 2023). While domain shift is not to be ignored between synthetic and real inputs (Sagers et al., 2023), an interleaved approach of real clinical queries and vetted synthetic surrogates provides quick progress towards balanced decision boundaries without oversaturating expensive dermatologist workloads early in development cycles.

In short, fairness-aware active learning with a tight integration with SSL debiasing techniques (e.g. RSMATCH filtering) and domain-adaptive pref-mapping is a cheap but clinically reasonable labeling for dermatology algorithms aimed in skin cancer detection from ISIC or HAM10000 (Raju et al., 2024) when demographically unbalanced data is used. The iterative refinement technique is simultaneously applied to sex and skin tone subgroups while respecting operational constraints to comply with privacy laws and realism of deployment, two of the primary parameters for ensuring sustainable translation of research prototyping into clinical trusted support devices.

2.4 QUALITY ASSURANCE IN SYNTHETIC DATA

2.4.1 ARTIFACT AND HALLUCINATION DETECTION

During the acquisition of synthetic dermatology images, detecting and reducing possible artifacts or hallucinations is critical in order to maintain both the clinical validity and the fairness goals of the augmentation proposals. GANs or diffusion models that add structural distortions, color banding, or implausible anatomical features may carry through with the semi-supervised learning loops into model decision boundaries, especially if they are more common in synthetic samples aimed at underrepresented subgroups like higher Fitzpatrick skin types (Xing et al., 2023).

Even small departures from plausible morphology may have huge consequences; for example, removing or mixing down pigment networks in darker skinned melanoma cohorts may result in artificially inflated classifier performance during training, yet poor sensitivity when used on real-world lesions from Fitzpatrick V-VI patients. This danger is compounded in SSL deaths where pseudo labeling could treat low-quality synthetic examples as equally valid as real-life instances unless filtered (Yamaguchi, 2023).

The first level of protection is in terms of the implementation of modules for automated detection of artifacts directly in the CHAIN of data generation. One subfield of adaptive methods is to train convolutional classifiers or statistical anomaly detectors on curated collections of "clean" versus artifact-bearing images, and to score the results and reject them if the plausibility is below a pre-training threshold (Georges-Filteau & Cirillo, 2020). Local texture clustering is very often used as a metric for these gating steps, because local anomalies such as checkerboard, ring artefacts or unnatural shadowing are very early signs of generator instability. Patch-based discriminators originally used in GAN architectures to enhance microstructural realism can be retroactively used as automated critics assessing small receptive fields for medical plausibility instead of global aesthetic quality. In effect, class-specific discriminators (e.g., melanoma-centric) are more sensitive to lesion-relevant aberrations than generic critics and therefore, the level of detection sensitivity is consistent with the clinically relevant structure important for diagnosis (Khader, 2023).

Diffusion-generated samples tend to be less prone to high frequency artifacts than early GANs, but hallucinations in the base dataset when rare morphological variants are missing from the conditioning prompt or when target subpopulations have been sparsely represented in the base dataset (Sagers et al., 2022). Hallucinations that are included include biologically impossible lesion morphologies, the background-skin tone transition, and illogical internal distribution of pigmentations. These errors may be methodological in nature, such as excessive smoothing that results from the use of an overly aggressive denoising step-schedule that destroys diagnostically relevant

irregularity. This needs to be resolved by adding structured semantic constraints such as lesion segmentation maps to direct image completion to clinically meaningful shapes (Georges-Filteau & Cirillo, 2020). Compensatory conditioning paradigms have been used and are effective in other medical regions in maintaining anatomic integrity within selected regions while maintaining variability outside regions; use in dermatology would reduce hallucinations at the source.

Though automation has an important role for scalability and economization, expert dermatologists also ensure a critical quality assurance gate, for example, for batches for sensitive areas of coverage as Fitzpatrick VI melanomas where underdiagnosis may have the greatest stakes (Pakzad et al., 2022). However, it is costly to employ expert time; thus hybrid solutions, where algorithmic pre-screening of synthetics is followed by selected expert review of borderline-scored ones, are both cost-effective as well as quality conscious. Similarly, the active learning principles used in annotation workflows may be used here: active learning samples are prioritized for expert annotation (e.g. samples with detection scores closest to decision thresholds), as their classification is more uncertain in terms of downstream fairness and ROC-AUC parity (Sagers et al., 2023).

Further, we show that gating can be made more robust by incorporating multimodal quality estimates. Objective similarity indices (PSNR, SSIM) when compared to threshold sets of quality real images provide quantitative information, but have to be interpreted carefully as high SSIM does not guarantee clinical plausibility when unlikely color or structures appear (Chang et al., 2023). Expand this to quantify image fidelity using FID and its functionally equivalent distance variant Demis, Dist-FID, that uses differences between synthetic and real datasets learned in the same feature space; monitoring variant behavior over training epochs identifies diverging behavior that correlates to hallucination risk mid-reality generation cycle. Empirical results show that using a combination of perception-based metrics and task-aware utility checks (i.e. assessing classifier performance trained on candidate synthetic batches against held-out real) is found to correlate better with true downstream impact when compared to each respective metric (Xing et al., 2023). Furthermore, utility alignment of image quality assessment to deployment needs directly enables principled cost-effective pruning: only synthetics shown to improve task-specific subgroup-specific ROC-AUC or calibration and not to degrade majority group accuracy are kept.

An important risk factor that has not been studied is systematic subgroup imbalance in artifact frequency. Although analyses have been conducted in dermatological literature to gauge skin tones' extent of bias, each of the described generators may result in clean exact reconstructions for Fitzpatrick I-III and image noise or chromatic artifacts for IV-VI (Wang et al., 2022). Indeed, if not identified, these imbalanced imperfections can serve as another source of degraded performance on minority skin tones, not because there is insufficient exposure to minority faces in the model but because feature learning is misled by quality differences. In addition to such mainstream measures of fairness, dataset-level audit stratified by demographic metadata should report artifact prevalence per subgroup to identify such skew before being integrated into SSL pipelines.

Domain shift is effective detection: anomaly filters trained only on dermoscopic data from HAM10000 may not detect artifacts introduced in synthetic smartphone-like images that are effective in testing the deployment for realism (Choudhary et al., 2020). In order to maintain the sensitivity under different target environment, the detectors must undergo interdomain calibration, either by multi-source training across the different modalities of the unseen devices or by adapting the candidate images to a canonical feature space using domain adaptors such as layers before the scoring is performed. This is of importance especially when generators are asked to simulate acquisition condition diversity in addition to demographic variability for training classifiers for heterogeneous clinical environments.

Finally, integrating artifact/hallucination detection into MLOps pipelines provides long-term governance over changing relationships between generators in drift conditions. As base generative models are fine-tuned or prompt libraries are advanced (for instance through targeted skin-tone variability workflows), automated detectors paired with periodic human testing stand as sentinels against any degradation that makes its way into less-frequently-sampled strata. Differently from privacy compliance safeguards under GDPR/HIPAA that differ between plausible unrecognizable surrogates and risk-bearing memorized resemblance, the detection module becomes part of a much more extensive trust architecture in which representation balancing efforts don't backfire through poisoned inputs that leak bias/noise into production systems (Chang et al., 2023).

Plausibility pre-integration and coupling decisions and utility-aware fairness measurement at the subgroup level transform artifact and hallucination detection from a fringe protection to a core element of cost-efficient clinically believable dermatology AI pipelines maintaining equal accuracy distribution across the demographically all-encompassing targeted subgroups.

3. SEMI-SUPERVISED LEARNING ENVIRONMENT AND DEPLOYMENT WORKFLOW

3.1 CORE SSL ALGORITHMS

3.1.1 CONSISTENCY REGULARIZATION METHODS

Semi-supervised regularization methods are built on the fact that the predictions of a model should not change substantially if inputs to the model are perturbed in an allowable way, whether it be a spatial perturbation, temporal perturbation, or photometric perturbation. This assumption is especially true for dermatology datasets such as ISIC, HAM10000, BCN20000, and Fitzpatrick17k, where issues with chronic representation exist among Fitzpatrick skin tones and sexes.

By enforcing prediction invariance under controlled transformations, such methods can help to compensate for the lack of data in underrepresented subgroups without necessarily requiring large numbers of annotated samples. Such methods are not only consistent with fairness goals but cost-efficiency goals, as well, assuming limited annotation budgets and that diversity improvements in synthetic data are better than perfect shadow data.

With regards to realistic dermatology AI pipelines, consistency limitations can be laid out along many axes. Spatial consistency aims at enforcing the label agreement between original images and geometrically transformed versions (rotations, flips, mild scalings) with lesion integrity constraints. Since the morphology of lesions along with pigmentation patterns, in the form of variegated color patches, is amongst the most important features for diagnosis, particularly transformative propensity in an aggressive manner could complicate features that are important for the correct classification.

Temporal consistency can generalize the invariance checks over related instances that are acquired at different time points or under different clinical conditions; for instance, a patient diagnosed multiple times using different imaging machines (Mu et al., 2020). Finally, in melanoma follow-up settings where subtle changes in lesion phenotype over months may be explained by a high degree of prediction stability, which underlies the confidence in the clinician-based longitudinal surveillance.

Equivariance-based constraints introduce an additional nuance by assuming that changes in the predictions are expected to mirror the change that is made to the input (e.g. if an image is rotated, it is expected that the segmentation mask will also be rotated). Training in this way constrains the space of possible models, which reduces the danger of models drifting towards spurious correlations associated with bias in tone or the sex-ratio of patients rather than actual pathologies. For example, in the absence of constraint mechanisms, a network can take advantage of background skin tone shadows as shortcut paths to classification, which is not acceptable in fairness-sensitive scenarios where darker tones are under-represented (Wang et al., 2022).

Consistency regularization is shown to be very useful in pseudo-label reinforcement within SSL frameworks such as FixMatch and MixMatch. Specifically, by providing, through perturbation-consistent checks, an artificially generated label for unlabeled data, it is only kept if the model converges on it across multiple augmented views (Yamaguchi, 2023). This defense against contamination provided in active learning loops in Section 2.4.1, where low-quality synthetic images or biased seed predictions might catastrophically generalize error into minority groups. We further tighten the retention criteria by implementing an adaptation from RSMATCH that dynamically re-tunes the confidence threshold that is used to retain the instances into these subgroups by taking into account the error profile for each subgroup during training iterations.

From an operational perspective of dermatology AI development pipelines, combining spatial and temporal consistencies is easy but clinically powerful. The Pseudo-Label Generation (PL-Ge) method (Mu et al., 2020) identifies and links both these constraints together in an end-to-end estimations framework, where both the synthetic labeled source datasets (X_s, Y_s) and pseudo-labeled real target datasets $(X_t, \hat{Y}_t)$ are jointly trained under a KL-loss using weighted sum of squared errors in both spaces. Formally:

$$L^{(n)} = \sum_i \text{LMSE}(f^{(n)}(X_s^i), Y_s^i) + \gamma \sum_j \text{LMSE}(f^{(n)}(X_t^j), \hat{Y}_t^{(n-1),j})$$

With gamma controlling the balance between source-side supervision and target-side pseudo-supervision, adapting gamma per subgroup enables more weight can be put on consistency learning in demographic slices with

few labeled samples (e.g. Fitzpatrick VI). This concerns fairness and the optimal usage of computational efficiency: expensive augmentation cycles may be prioritized to the situations where parity gains are the highest.

Another long-standing threat is domain shift between synthetic source data and real target samples which exist when synthetics are generated from GAN or diffusion models that are imperfectly approximating tonal microstructures (Yamaguchi 2023). If the decision boundaries of models are highly localized to the synthetic artifacts rather than the more biologically meaningful pathological features, their robustness is shown to possibly decrease on actual high FST melanomas, even as their robustness on augmented pseudo-labels appears to be high. Cross-domain feature alignment (adversarial adaptation or embedding normalization) can be utilized before applying consistency checks in order to apply perturbations to clinically meaningful representations instead of artifacts in style representations.

Cost effectiveness analysis of these approaches places them competitively against fully supervised expansion strategies: once deployed within training loops, constraint enforcement executes as part of normal augmentation logic without the extra expert markup cost. Multiple forward passes per sample inside the post-transformation pass enable the computation overhead; however, when one carefully schedules each pass and constrains sets of perturbations to transformations empirically observed not to perturb the lesion geometry, the excess GPU overhead is diminished while still achieving high ROC-AUC gains across minority subgroups.

Stratified metrics continue to be monitored during training: statistically significant gains for all areas need to be validated for each Fitzpatrick and sex subgroup: ROC-AUC retains meaning across the sex/skintone groups at the same specificity (e.g., 90%), PR-AUC curve showing the distribution of lesion rarity per group, and calibration plots showing probability retains meaning.

Real-world deployments favored beyond the lab evaluation for complexity. Whereas the clinical acquisition heterogeneity among patients, the camera type heterogeneity between hospitals, the dermoscopic lighting rig vs. smartphone macro lens representation heterogeneity all suggest that perturbations used during the consistency regularization should represent such variability. The use of simulated instrument-specific noise patterns or color changes allows the models to train with the invariance to operational deployment-relevant changes without collecting years of multi-site measurements. Too close to the line and you may end up with unrealistic distributions that are detrimental to the generalization and less trust from you if you use the calibration curves (drifts) against certain acquisition conditions after deployment.

Incorporating consistency does not make sense in itself but can be treated as a modular component with adjustable subgroup effect, audited from time-to-time while tracking fairness metrics based on unlabeled pool incorporation and active learning selection patterns. Pipeline governance layers make downstream outputs clinically plausible for any segment of the target demographics, while making transparent any changes made mid-course due to drift such as recalibrating g for changing demographics under GDPR/HIPAA audit requirements (Thong et al., 2023).

By combining the spatially-temporally equivariance constraints with robust pseudo-label filtering in the context of SSL and real and synthetic dermatological images, these techniques contribute directly to representation gap closing and reduced annotation cost. However, their success relies heavily on domain-awareness in augmentation design, subgroup imbalance profile-specific hyperparameter adaptation, continuous metric stratification over train-monitor operations, and operation-readiness for clinical capture in diverse real-world deployments for cost-sensitive reliability in fairness-driven diagnostics.

3.1.2 FIXMATCH AND MIXMATCH ADAPTATIONS

As two popular SSL paradigms for computer vision tasks, FixMatch and MixMatch are able to leverage huge amounts unlabeled data with pseudo-labeling and augmentation-based consistency regularization. Dermatology AI workflows based on ISIC, HAM10000, BCN20000 and Fitzpatrick17k need to be carefully adapted to specific fairness objectives across Fitzpatrick skin tones and sex groups, annotate and compute-budgets limits and be robust to deployments' device quality and acquisition protocols.

These methods' fundamental dependence on augmentations as both pupal labels and strengtheners of pseudo-labels provides a natural interface with synthetic data generation pipelines described in Section 3.1.1; however, and also adds to contamination risks in the presence of initial modeled biases (Yamaguchi, 2023).

In conventional FixMatch, strongly augmented images (as augmented from low resolution counterpart) with high quality confidence scores (above a threshold τ , typically 0.95 in many cases) from the weakly augmented images

are labeled as hard pseudo-labels to their corresponding images. Due to limited quantity of minority-skin-tone lesions, the softmax confidences of minority-skin-tone lesions tend to be lower than majority-skin-tone lesions, posing a problem of low-threshold adaptation issue in imbalanced dermatology datasets, where some fixed threshold may bias to majority-skin-type classes.

To reduce this imbalance without compromising clinically meaningful decision thresholds, subgroup-aware dynamic thresholds can be employed, in which τ is selectively reduced for Fitzpatrick IV-VI strata or underrepresented sex categories. This means that more pseudo-labeled samples will be taken into account in the loss calculation, which is beneficial for minorities to improve the loss calculation when dominated classes have higher t values than minorities to prevent the amplification of wrong predictions (Wang et al., 2022).

For filtering, RSMATCH-based subgroup-aware multi-view filtering further eases the propagation of mislabels into sensitive slices for fairness in a way that agrees with subgroup-conditioned confidence envelopes, by adaptively filtering out pseudo-labels whose agreement across multi-view perturbations is low (Yamaguchi 2023).

In addition, it is shown that the averaged approach of MixMatch - that first augmentations together, and then sharpen various averaged predictions before mixing labeled and unlabeled samples with MixUp - can also be adjusted for equitable-sensitive settings. When synthetic common skin tone images produced using GAN or diffusion modeling are included in unlabeled pool, the label sharpening has to consider the possible synthetic-to-real domain gap (Chang et al., 2023).

Before prediction averaging, it is recommended to apply domain alignment steps (e.g. feature normalization of real/synthetic domains), in order to avoid that the augmentations applied in the domain with the real data are biased towards modes of majority-domain. Furthermore, the linear blending implied by MixUp might result in impossible skin tone blends between pairs of very different skin tones; limiting MixUp partners to reside within a set color-space distance (such as ΔL^* values computed from CIELAB coordinates) maintains clinical plausibility while still gaining regularization effects (Thong et al., 2023).

In both algorithms, optimization of the augmentation policy is at the core. Dermatological-specific augmentations need to reflect realistic clinic-to-clinic variation, gentle geometric transformations, photometric transformations reflecting variation in dermoscopic illumination, fine resolution degradation, but should not morphologically distort lesion features important for malignancy detection. Imposing device-specific noise models (smartphone macro vs. dermatoscope) to the robust augmentation stages, thus preparing the models for deployment in heterogeneous settings, while preserving the true structure of the lesion (Mu et al., 2020).

Overzealous color or contrast alterations risk the simulation of pigmentation patterns which do not occur physiologically in some Fitzpatrick types leading to degraded subgroup ROC-AUC despite potentially high aggregate accuracy post-training (Wang et al., 2022).

FixMatch/MixMatch is popular for bridging synthetic data implementation into clinical workflows, as it means synthetic data do not have to be shared between centers and our patients are not identifiable through images, but there has to be a new dimension on quality assurance (Yamaguchi, 2023). Ghastly, low-quality synthetics and hallucinations that are particularly skewed towards darker hues may inadvertently (and counterproductively) bias features from that same demographic (Xing et al., 2023).

However, pseudo-labels are only assigned if candidate images pass plausibility thresholds which can be tuned per class/Fitzpatrick type (Chang et al., 2023) before unlabeled batches enter SSL pipelines using artifact detection scores as gating variables. This gating is well matched to confidence filtering by eliminating any lesions that, despite their predicted probability magnitude, are unacceptable to the user based on other structural reasons.

There are cost-efficiency considerations that will limit the amount through which augmentation cycles can be scaled through the training epochs. Both FixMatch and MixMatch have compute overhead caused by multiple forward passes over augmented views with different augmentations, and dynamically adjusting the number of strong augmentations per sample using trends in subgroup performances can be used to reduce resource consumption without compromising parity gains.

For example, when there is convergence of ROC-AUC between Fitzpatrick II and VI melanomas after a few epochs, then additional costly over-sampling for type VI cases can be reduced while maintaining baseline fairness levels measured at sensitivity-at-90% specificity thresholds (Sagers et al., 2023).

Clinical credibility: More than the overall accuracy, it should be demonstrated that these subgroup-specific model calibrations (SLAs) yield equal improvement of calibration in subgroups, so that the predicted malignancy probability corresponds well with the actual likelihood of malignancy (Thong et al., 2023). Divergence may be seen early as it is periodically checked as part of the training program for calibration; otherwise, confidence in fewer minority types and false positives may induce unwarranted biopsies or undetects even if superficially higher sensitivity rates are achieved. Probability outputs can be stabilized without sacrificing needed classification acuity for rare lesion identification, when only probability-aware loss adjustments are made (e.g., temperature scaling within subgroups during SSL training).

To alleviate domain shift between curated training corpora such as HAM10000 and images from the field, co-training with an auxiliary task where it is available (e.g., lesion segmentation mask prediction, when masks are available) forces FixMatch or MixMatch to maintain shared feature representations that are not susceptible to capture-condition noise (Mu et al., 2020). Such multi-task alignment allows to distinguish pathologic features from tone/background artifacts, which is a feature of crucial importance when aiming towards fair across phenotypes, not only characterized by different tones, but also textural contrast profiles depending on melanin concentration.

Finally, the integration of these SSL variants into MLOps-managed workflows assures that the cost-benefit ratios are consistently monitored over time. Continuous integration of live drift monitoring of demographic distributions within incoming unlabeled streams under GDPR/HIPAA logging restrictions can also be utilized to update dynamic thresholds or hone parameters when evaluating new streams with emerging properties (Choudhary et al., 2020). Automated retraining scripts can trigger context-aware FixMatch/MixMatch hyperparameterization preferences rather than manual human delays by prioritizing fairness fixes when the pool that has just been ingested has changed to be heavily underrepresented (e.g. a surge of type V-VI cases from a new clinic).

These changes can help deliver both equity and cost-effectiveness gains in combination with rigorous synthetic data QC pipelines and stratified validation regimes based on ROC-AUC, PR-AUC, sensitivity-at-90% specificity, and calibration parity reporting per PA slice, all while being operationally feasible in real-world clinical safety-mandated dermatological AI implementation.

3.2 INTEGRATING SYNTHETIC DATA INTO SSL

3.2.1 MIXING POLICIES AND RATIO SCHEDULING

Mixing policies and ratio scheduling in the context of incorporating synthetic dermatological data into semi-supervised learning models such as FixMatch and MixMatch needs to be carefully balanced to achieve the goals of fairness, cost-effectiveness, and clinical applicability. The basic choice is how synthetic and real sample are mixed during training, rather than at the global level, but also at the level of particular groups of demographics for example Fitzpatrick IV-VI Skins or underrepresented sexes.

If the primary objective of synthetic data is to increase minority group representation, the policy of mixing must be designed in a way that ensures that the quality of its integration lifts subgroup-specific metrics of the data, ROC-AUC, PR-AUC, sensitivity-at-90 per cent specificity rather than raising the global average while preserving disparities (Sagers et al., 2023). A fundamental risk is of synthetic-to-real domain shift, mainly this can be acute when generated samples in darker skin tones lack microstructures such as pigment networks that can be familiar to clinicians, and are well represented in real datasets for instance HAM10000 or ISIC archives (Choudhary et al., 2020). A naive mixed-batch strategy, in which all sources are given equal weight, can lead towards model-learned representations drifting toward the artifacts that are prevalent in synthetic subsets, and which are not clinically meaningful.

Ratio scheduling helps to mitigate this issue by dynamically varying the ratio of synthetic-to-real samples with training epochs. A common pattern is to front load synthetic data early in SSL cycles, where there are still few labeled real minority group cases, and then tapering down its contribution as model exposure to verified real minority subgroup images are increased in an active learning and targeted annotation loop (Pakzad et al., 2022). This method prevents the over-fitting of decision boundaries in the early stages and controls the contamination when the confidence of the pseudo labels reaches the stability thresholds for minority types, as used in the RSMATCH filtering (Yamaguchi, 2023).

The mixing policy itself can be either per-batch mixing or per-epoch mixing. Per-batch mixing creates synthetic diversity in each gradient update, which attains maximum fairness per iteration at the cost of being too exposed

to domain mismatch noise if filters are too weak for quality assurance (Xing et al., 2023). Conversely, per epoch segmentation, devoting some epochs fully to high quality synthetics, then exclusively real data epochs, can mitigate against intra-iteration domain noise but include a risk of catastrophic forgetting for rare subgroups unless replay buffers sustain their representation between transitions. In dermatology contexts where clinician trust is paramount and calibration parity between skin tones strata is paramount (Thong et al., 2023) hybrid approaches of batches skewed toward synthetics for minorities, together with backbone weights stabilized via continual small doses of majority real cases prevent shift both towards style artifacts and subgroup imbalance reinforcement.

Dynamic ratio scheduling takes advantage of feedback loops that are directly related to fairness measures that are monitored during training. For example, if we observe ROC-AUC improvements at a level plateau for Fitzpatrick VI melanomas but still below parity goals with type II melanomas, scheduling algorithms can target a higher percentage of type VI-targeted synthetics within We can sample the next batch as mixes until parity gains are observed (Wang et al., 2022). This necessitates fine-grained batch composition controls that are able to bind demographics metadata in conjunction with source-type labels (real/synthetic) in order for both mixing proportions and augmentation pipelines to be subgroup-aware.

Cost efficiency means not just keeping track of the performance, but also GPU hours for high fidelity generation (diffusion models tend to be much more compute expensive than GANs, and adjust proportion goals appropriately when a loss of ROI is identified (Khader, 2023). SSL contamination prevention is an integral part of developing scheduling logic. If the level of agreement between pseudo-label assignments for synthetic images is less per strong augmentations than the level of agreement between pseudo-label assignments for equivalent real images, particularly in underrepresented tones, then use the mixing system to automatically de-prioritize such samples or route them via secondary label refinement before reintegration into main training streams. The combination of automated artifact/hallucination detection scores into ratio decisions further ensures that unsound feature-space geometry does not get degraded with implausible lesion inputs focused on sensitive gaps such as type V-VI representation (Chang et al., 2023). The scheduler can force minimum quality at the batch assembly stage: batches that do not pass demographic-specific plausibility checks skip to inclusion until regeneration passes through QC gates defined in previous sections of the generation.

Another aspect to be considered is augmentation-intensity asymmetry between real and synthetic inputs for mixed policies. Real minority group images might have a fairly natural variation from various capture devices than augmentation might biologically, hence again light augmentation might work. Synthetic outputs on the other hand have the advantage of having better clinic-simulation augmentations (lighting changes, mild resolution degradation) to prepare them for the world of deployment realism outside controlled generation pipelines (Sinha et al., 2024). Ratio schedules thus require linked augmentation profiles for the source type/demographic pairing: GAN/diffusion-based synthetics for dark skin tones may be richly variably injected early-on to simulate captured diversity missing from base generation runs but gradually approach augmentation intensity matching equivalent real cases as ratios make a ratio approach at deployment-ready blends.

An operationally resilient scheduler also supports MLOps governance requirements under GDPR/HIPAA by providing logs on every batch composition, as well as ratio shifts performed due to fairness profile feedback (Thong et al., 2023). Such traceability is supportive of regulatory reviews, demonstrating the equity approach in demographics used to make concrete adjustments during model development cycles. Moreover, there are long-term maintenance benefits (ratio targets can dynamically adapt according to incoming unlabeled pool composition shifts, e.g. a spike in Fitzpatrick IV-V clinic submissions triggers recalibration upwards in their synthetic supplementation until active learning review pipelines catch up with strong label strongness from dermatologist audits).

Cost-control incorporation on the ratio planning inside the plan is grounded with a quantitation of marginal utility per additional synthetic sample consumed: the correlation analysis among marginal increases in fairness metric by the number of synthetics injected is helpful set the best saturation points beyond which extra synthesis produces very little in gain on compute-cost (Sagers et al., 2023). After these points have been achieved for certain demographics or lesion types, using scheduler logic ratios can be frozen at baseline maintenance levels instead of continuing costly cycles of generation without an obvious return. A similar principle operates the other way around, when active learning campaigns provide new labelled minority group real data, synthetic ratios adjust downwards proportionately to reflect an improved genuine coverage without being fair to improvement benchmarks achieved so far.

Finally, clinical credibility requires validation of selected mixing policies that do not distort calibration curves across demographics if evaluated at fixed specificity settings that are critical to triage workflows. Even if there is

a uniform improvement in sensitivity-at-90% specificity after balanced-ratio scheduling with targeted synthetics, misaligned biopsy referral priorities in spite of apparent raw sensitivity parity may arise due to misaligned calibration slopes between skin tone groups (Thong et al., 2023). Consequently, ratio schedulers should integrate allocation changes with calibration-aware loss tuning steps to ensure that probability output consistency is not broken between all the phenotypes being represented during training updates based on demographic-targeted performance feedback loops.

By incorporating awareness of demographic metadata (in both per batch composition rules and epoch-level ratio trajectories) and gating the entire process of composition rules and epoch-level ratio trajectories through artifact detection scores, synthetic dermatology contamination filters, cost performance analytics, and fairness-calibrated validation metrics, it's possible to incorporate synthetic dermatology imagery into SSL pipelines without compromising generalization capacity and compliance readiness while effectively reducing the cost of annotation (tied to chronic gaps in representation across Fitzpatrick tones and sex subgroups targeted for improvement) (Sinha et al., 2024).

3.3 MLOPS CONSIDERATIONS FOR CLINICAL DEPLOYMENT

3.3.1 MODEL REGISTRY AND VERSION CONTROL

The development of effective semi-supervised learning architectures using synthetic dermatology data will be translated into a clinically acceptable system with effective model registry and fine tuned version control for a cost-effective implementation.

The complexity of the multiple sets of data and images involved (ISIC, HAM10000, BCN20000, Fitzpatrick17k, GAN- or diffusion-generated imagery) renders storing provenance, configuration parameters and performance metrics not only a technical defense but also a fairness-critical task.

The registry serves as the central reference library of all model artifacts and contains links to training datasets that include explicit representation metadata such as distribution on Fitzpatrick skin tones and sex subgroups.

In the absence of these audited connections, monitoring of equity-based models is opaque, in the sense that it is impossible to confirm that the performance parity obtained for a validation process is also reflected in the updated models (Pakaszad et al., 2022).

Versioning: Typically developed in code commit, version numbering needs to extend to data set snapshots, model weight histories, hyper parameter settings, augmentation policies and even SSL mixing ratios over time.

Registered versions should hold demographic-stratified ROC-AUC values, PR-AUC scores, sensitivity at 90% specificity by Fitzpatrick type and sex, and subgroup-specific calibration curves, meaning that regressions in fairness are identified by downstream analysis as readily as global accuracy declines (Sagers et al., 2023).

Similarly, quality assurance indicators on synthetic training inputs should also be included: percentage of synthetic samples that can pass artifact/hallucination detection filters within each subgroup (Chang et al., 2023) and any domain adaptation steps used prior to its integration into SSL pipelines should be recorded.

Given the risk of synthetic-to-real domain shift in minority-tone lesion samples (Choudhary et al., 2020), each registry record should record source type (real vs synthetic), generator type (GAN or diffusion), conditioning parameters (e.g., Fitzpatrick targets from prompt engineering), augmentation levels after generation, artifact removal process (Khader, 2023).

This metadata enables version auditors to identify reasons if fairness metrics deteriorate unexpectedly after a new synthetic batch has been introduced, either because of style drift due to generator fine-tuning or because of contamination by low-confidence pseudo-labels (Yamaguchi, 2023).

In these circumstances, version control rollback is an essential feature to allow going back to the previous stable check points without losing historical equity gains. Complete lineage records also help to support compliance reporting under GDPR/HIPAA as they provide explainable justification for representation changes of protected attributes through successive updates (Thong et al., 2023).

It is also where cost-efficiency considerations overlap: version records can contain compute cost per training run and GPU-hours per epoch for each loop of training with combination of real and synthetic examples, as well as annotation cost per labeled case obtained by active learning per training cycle (Abir et al., 2024).

By measuring the change in performance for these beliefs versus these expenditures across versions, teams will be able to tell if there are diminishing returns from certain augment or labeling strategies. For example, one might find in future versions that increasing synthetic dark-skin representation via expensive diffusion generation obtains little parity benefit but considerable compute overhead compared to lower-cost GAN augmentations vetted with artifact filters (Georges-Filteau & Cirillo, 2020), in which case ratio schedulers will be adjusted.

The registry should also include artefacts (bias audits, output logs from RSMATCH pseudo-label filtering runs with details on acceptance/rejection rates per subgroup etc) to provide transparency on contamination risk over time (Yamaguchi, 2023). These artifacts can subsequently be used not only to perform technical diagnosis for scenarios of skew on minority class confidence distributions but as an evidentiary platform in form of actual data demonstrating that proactive mitigation action was applied in every model iteration. Coupling these with diagnostics of calibration conditionalizes whether increased sensitivity was at the cost of misalignment of its likelihood with the demographic classes present. Operational design should easily account for heterogeneous deployment environments: models trained on dermoscopic images risk being challenged by clinic settings with a variety of factors including light spectrums for smartphone macro lenses.

Adding modality tags to dataset descriptors that are associated with individual registry entries allows for domain shifts either by adding coverage to new acquisition types or augmentations that have been chosen to mimic such shifts to be tracked over time with their measured impact on generalization factors (Baugh et al., 2023).

Deployment realism objectives are always achievable, and if one aspect of the environment in a new release is not aligned with the others because of an unaccounted for modality drift, there are means of reverting back with full context of the environment, intact. The registry-versioning loop can be automated by monitored fair parameters. For example, if a new model graduates a sensitivity-at-90% specificity for Fitzpatrick VI melanoma below a clinical threshold established at majority-type baselines after an update (Wang et al., 2022), it is possible to immediately run retraining jobs seeded from previous stable versions while stopping deployments of the failing model in production.

Such triggers implement a time to equity risk exposure windows, which is an important safety mechanism for which the implications for morbidity are severe if not detected in a timely manner. Structured registries also assist in enabling institutions to collaborate in order to scale without sacrificing privacy commitments. When exchanging weights across sites under stringent data protection settings, appending metadata to the linked dataset with only distribution statistics (skin tone/sex proportions) (as opposed to the raw patient images) makes cross-site replication of fairness audits possible (without breaching GDPR/HIPAA). These version control systems can result in immutable logs for any change of training configurations or input compositions, which can prevent silent drift to imbalanced states during autonomous local fine-tunes. Finally, integrating specific features of artifact/hallucination frequency analysis into the registry, ensures its performance in the long term because of the slow degradation of generative sources with time (Khader, 2023).

As generators are upgraded or prompt libraries change, in the work that we mention above for skin-tone changes, the frequency of passes vs. failures of proposed outputs for plausibility may change minimally; without baselines per generator version recorded in the registry, we miss early warning signals. Increased transparency: A direct link to version histories allows us to see such drifts in ongoing comparative dashboards, before they are amplified by post-deployment bias amplification.

Hence, the combination of an extremely strictly maintained model registry and full-fledged version control not only strengthens the credibility of the model in the eyes of the clinic but also became the operational cornerstone for balancing cost-effectiveness with continued equity with respect to various patient populations. It makes every abstract equity promise into measurable technical practice: every transformation recorded, the delta on minority-subgroup performance and global accuracy, every ingredient from the source (generation) to the destination (SSL integration & context of deployment), all out in the open for clinicians, ML engineers, and regulators.

3.3.2 PERFORMANCE DRIFT AND FAIRNESS MONITORING

To detect drifts in AI systems in dermatology towards certain groups, it is necessary to track aggregated data and subgroup-level results over long periods of time in a systematic way. This involves designing workflows

which capture change in predictive behaviour with time, whether due to changes in distribution of input data, parameter changes in model retraining or changes due to demographic composition of real world clinic submissions. This approach can be further integrated with registry and version control processes as detailed above to trace all measured drifts to their root causes and fix them in a transparent manner. A main reason for drift is distribution change of the data, also known as data drift (or covariate shift).

In clinical dermatology pipelines generated from ISIC, HAM10000, BCN20000 and Fitzpatrick17K, there can be changes when new acquisition devices introduce subtle changes to colour balance or resolution (e.g. replacing dermatoscopes with smartphone macro lenses) or when seasonal or geographic differences change the prevalence of some lesion types (Kore et al., 2024).

Because under-representation already exists in darker Fitzpatrick skin types, small changes in case mix can disproportionately impact minority-subgroup performance; with recent patient cohorts that are skewed towards lighter skin tones, model adaptation may actually optimize away rare-tonal structures pertaining to higher FSTs. Instead of taking global accuracy changes as suitable proxies, automated monitoring consequently has to evaluate drift symmetrically on all subgroups. As a result, in order to instrument such systems, continuous evaluation pipelines can be applied to sample model outputs against a fixed gold-standard validation set representative of the intended deployment population. This range must be broken down by Fitzpatrick type and sex to allow sensitivity at 90% specificity thresholds per sub-group to be observed over time (Sagers P, et al., 2023).

When these stratified quantities deteriorate below predefined thresholds, as is the case for clinically sensitive classes (e.g. type VI melanoma), alerts are raised for further analysis of source causes: a shift in demographics, an artifact contamination wave on injected synthetic augmentation (Chang et al., 2023), or a subtle slip on parameters after SSL loops with low-confidence pseudo-labels (Yamaguchi, 2023).

Technical approaches of detection range from simple comparison of current value of the metric with historical baselines to more sophisticated statistical tests for subgroup trend line significance. The use of compact multivariate (MV) methods additionally using the individual constancy (L^*) and chroma (h^*) verbalization enables the drift perception model to account for subtle tonal adjustments beyond pole shifts in categorical Fitzpatrick (Thong et al., 2023). For example, an increase in the amount of yellowish undercurrent in darker FST images by possibly a regional dataset contribution, can change lesion-background contrast ratios enough to diminish classifier sensitivity without any change in raw FST counts. Running the hue/lightness histogram in addition to monitoring ROC-AUC trends allows for prompt correction before rates of clinical error rise. Besides input attribution analysis, post hoc explainability methods have been investigated for drift diagnosis of internal reasons.

For medical imaging on high-dimensional data, however, caution should be exercised as several statistical critiques exist regarding the reliability of techniques such as SHAP values for providing explanations at the feature-level under particular assumptions. More focused auditing could compare saliency map patterns across time per lesion type per subgroup: if the model-detected regions drift away from patho-relevant regions to background noise for Fitzpatrick IV-VI after fitting fresh synthetic patch, it is indicative of a fairness-drift (and needs retraining further on stronger artefact/hallucination filtering). It is important to account for the inequity indications given by past studies that detection algorithms may additionally exhibit increased sensitivity fluctuation based on demographics (Kore et al., 2024).

For dermatology SSL workflows that combine mixing policies with ratio scheduling (Sinha et al., 2024), fairness-aware drift checks examine not only the initial parity at deployment but also temporal consistency; a balanced performance today might degrade if tomorrow unlabeled pool composition drifts. So feed forward figures from the active learning query log should be examined alongside inference cohort demographics to try to identify if annotation priorities have unintentionally led to a deviation of resourcing away from lower-performing protected groups. Fairness drift monitoring is linked with economic sustainability through cost-aware design. It is clear that absent a change in pipeline architecture, allowing training of more patients in a given period, GPUs wasted in repeated retraining with statistically insignificant changes in clinical performance will prevent achieving efficiency targets.

Correlation analysis between subject-specific ROC-AUC degradation and estimated risk of morbidity provides a ranking of intervention urgency: Acute retraining needed for degradations affecting melanoma sensitivity on minority tones with lower PR-AUC degradations on benign lesion classes being given lower priority. Operationalization into MLOps environments means such events as threshold events trigger automated retraining of the model as a matter of acceptable slope and divergence of calibration curves between majority and minority

groups, or RSMATCH rejection rates that climb above norms for certain demographics (Yamaguchi, 2023). These triggers are to be paced by governance controls in order for the interventions to be transparent under GDPR/HIPAA regulation; the demographic stratified drift snapshots in the model registry provide the ability for external review to ensure safety and equity commitments are being met. An additional advantage in utilization of drift assessment is the combination of input data monitoring and output distribution analysis.

If the score distributions move in the direction of overconfident predictions on darker-toned benign lesions in months dominated by synthetic GAN training updates (Georges-Filteau and Cirillo 2020) then calibration parity checks will show alertness to an overtriage risk - a fairness issue even if raw sensitivity rises. To improve such imbalances, SSL augmentation intensities per source species might be adjusted or ratio rates moved away from synthetic-intensive mixes until recovery in calibration (Wang et al., 2022). Finally, domain adaptation tools are used for preventive purposes against fairness-related drift caused when deployment modalities change (e.g., a hospital network has clinics with different imaging hardware).

Alternatively, in order to avoid abrupt accuracy change between demographic slices due not to actual feature sparsity but rather to mismatches in image styles, models normalize the incoming image styles toward baselines in the training domain through the adaptation layers before inference scoring, as Choudhary et al. (2020) suggested.

Used together with multi-metric subgroup-specific audit dashboards visualizing trends, ROC-AUC, PR-AUC, sensitivity-at-90%, calibration slope, etc., these adaptive controls maintain both equity and cost-effectiveness across evolving clinical landscapes while satisfying regulatory safety-critical AI system requirements over diverse patient populations.

4. VALIDATION AND COST-PERFORMANCE ASSESSMENT

4.1 EVALUATION METRICS FOR SKIN CANCER DETECTION

4.1.1 ROC-AUC AND PR-AUC

ROC-AUC and PR-AUC can be used as complementary but different measures of evaluating the performance of a classifier in tasks of detecting skin cancer, particularly in the fairness-sensitive conditions of representations gaps among datasets of ISIC, HAM10000, BCN20000, and Fitzpatrick17k. The ROC-AUC (Receiver Operating Characteristic - Area Under Curve) is a summary of the sensitivity (true positive rate) versus the specificity (1 - false positive rate) of all threshold settings. ROC-AUC can provide a threshold-free perspective on the discriminative ability and stratified validation per Fitzpatrick type and sex to uncover its subgroup differences (Sagers et al., 2023). This stratification is severe; world ROC-AUC can obscure the decreased sensitivity to non-majority skin tones in case unequal training corpora have tilted the decision-making space toward the appearances of the majority (Wang et al., 2022).

The shape of the ROC curve is a measure of how tolerant a model is to the false positives versus the true positives, which is closely intertwined with deployment realism as the urgency of a biopsy depends on predictive certainty. The presence of high ROC-AUC in Fitzpatrick I-III and significantly lower in IV-VI is an indication of the imbalance in feature generalization that may be translated to false melanoma diagnosis in darker skin groups. There are other risks associated with integrating synthetic GAN or diffusion output to training: synthetic-to-real domain shift can spuriously increase ROC-AUC by increasing the easy classification of synthetics prone to artifacts, and without real-world subgroup improvements (Choudhary et al., 2020).

This necessitates the integration of ROC-AUC interpretation with subgroup-specific assessment upon filtering generated imagery based on artifact/hallucination detection protocols (Chang et al., 2023). PR-AUC (Precision-Recall Area Under Curve), though, proves especially informative in situations in dermatology where the positive class, malignant lesions, can be a small portion of the data. As precision is a punitive measure of false positive, and recall is a rating of true positive capture, PR-AUC is more directly indicative of behavior when the model is operating in a skewed prevalence than ROC-AUC (Sheffield, 2023).

Class imbalance exacerbated by underrepresenting some types of lesions in larger FSTs, PR-AUC gives respite regarding whether augmentations or SSL-produced pseudo-labels can maintain the precision and increase the recall of the rare malignant events in minority subgroups. Synthetic contamination all of the accuracy drops (e.g., mislabeled GAN outputs) can negatively affect PR-AUC although ROC metrics are on the increase (Yamaguchi, 2023).

RSMATCH-style filtering addresses this risk by introducing confidence-enhanced pseudo-label selection criteria on a slice-by-slice basis before being used to serve the goals of SSL. Fairnesswise, besides reporting both metrics at the aggregate level of the dataset, it must also be disaggregated by Fitzpatrick skin tone and sex strata. An example is the sensitivity-at-90 -percent specificity point, which is a clinically relevant point of the ROC curve, paired with the PR values of each subgroup to show parity in the operational ranges most consistent with biopsy referral thresholds (Sagers et al., 2023).

Such multi-metric subgroup reporting is increasingly favored by regulatory requirements under GDPR/HIPAA as it helps to confirm fair results on the measured attributes (Thong et al., 2023). This reporting chain includes calibrated probability output, bad calibration may give inflated PR-AUC when the predicted probability distributions are biased to binary extremes, and the distributions do not correspond to the actual likelihoods within subgroups. Computational efficiency is also included in metric collection routines when in MLOps pipeline models are retrained on a common basis in response to drift events described above. Any selection of a subgroup-stratified ROC and PR curve can then be logged to enable traceability, as model versions across the version controlled registries which include additions such as ratio scheduling adjustments or augmentation policy updates can be compared (Pakzad et al., 2022).

This connection assists in the diagnosis of regressions of synthetic source shifts or new MixMatch augmentation sequences without performing deep forensic examination each cycle. Teams can optimize resource usage about cost-effectiveness by relating compute/gpu-hours on various conditions of augmentation and their influence in subgroup-specific gains of AUC (Abir et al., 2024). The increased sensitivity of PR-AUC to changes in precision means it can be useful in active learning cycles targeting minority groups: querying high-FST strata with uncertain cases, can increase the recall significantly, but decrease the precision in case the minority group has a higher

proportion of such cases returning benign diagnoses on expert review. Measuring PR depreciation in relation to ROC stability can be used to determine whether the active acquisition will be beneficial to the overall equity or it will also risk overtriage in deployment situations (Wang et al., 2022).

On the other hand, minimal or increasing PR and aligned ROC lifts have positive implications of sustained investment in specific annotation campaigns with minor changes in operational cost. In case domain adaptation strategies, translation of styles between the dermoscopic image and smartphone macro capture, are utilized to train models to work in the heterogeneous field condition (Choudhary et al., 2020), the metric stability in both directions acts as an extra layer of validation. This persistent decrease in PR-AUC after adaptation in frequency of dark-toned subsets may be an indication that adaptation eliminated clinically salient contrast cues related to pigmentation patterns; meanwhile, PR scores were higher in all modalities, which may imply that adaptation could not remove diagnostic signal. The combination of the two metrics in hybrid evaluation thereby provides a multi-perspective protection against the unnoticed performance erosion due to harmonization steps of styles or due to gradual drift of incoming case demographics over time (Kore et al., 2024).

ROC-AUC and PR-AUC both respond in the natural way to artifact/hallucination dynamics occurring in synthetic-heavy augmentation regimes. The prevalence of artifacts in particular tones might cause an artificial increase in recall by having easier-to-classify yet clinically non-significant features; this also negatively affects the accuracy as these features do not manifest in the real world. The combination of artifact prevalence audits on a slice-by-slice demographic basis with AUC calculation should also make sure that any fairness gains are attributable to lesion discrimination and not model shortcuts on biased synthetic sets (Chang et al., 2023).

These audits supplement domain shift analysis when synthetic generation is used (but not natural acquisitions, which are detailed in Section 3.3.2) and the metric gains are kept consistent between synthetic generation runs and actual clinical utility instead of statistical noise caused by controllable training artifacts. A well-substantiated combination of the two measures in the workflows of validation without necessarily prioritizing one over the other provides sensitive decisions in the course of the model refinement processes to ensure that the models address gaps in the representations without missing the resource allocations and legal requirements.

Using PR percentile change with per-subgroup ROC percentile ranks presents clinicians and auditors with a concurrent picture of operating characteristics both in balanced-threshold and imbalance sensitive situations that represent actual disease prevalence. This dual visibility eventually makes the trust stronger on cost-effective dermatology AI pipelines with fairness-aware augmentation, biases mitigation of the provided version, and MLOps-ready mechanisms of deployment that are adjusted to the long-term objectives of clinical credibility in diverse patient groups that may need improvement (Sagers et al., 2023; Sheffield, 2023).

4.1.2 CALIBRATION AND LIABILITY CURVES.

Calibration and reliability curves offer a very important and critical supplement to discriminative metrics in the process of assessing models of skin cancer detection in heterogeneous datasets that include both real and synthetic dermatology images. Whereas such measures as ROC-AUC and PR-AUC measure characterization of ranking capability or precision-recall trade-offs, calibration analysis is an inquiry into the content of whether the model predictive probabilities accurately mirror empirical outcome frequencies at all confidence levels. Miscalibration has a direct patient safety impact in clinical setting where probabilistic results represent a factor that might affect biopsy referral decisions or lesion surveillance: in benign tumours an overconfident model can cause unnecessary excisions, whereas in malignant lesion treatment an underconfident model would result in under-treatment.

This problem is exacerbated in fairness sensitive environments in which ISIC, HAM10000, BCN20000, Fitzpatrick17k, and other datasets have systematic representation biases by Fitzpatrick skin tones, sex. A model may seem to fit well on the aggregate level but the calibration error in the subsets that are underrepresented could be high. This is plotted using reliability curves, which counts predictions into confidence bands (e.g. 0.0-0.1, 0.1-0.2) and then graphically against the observed accuracy in each bin. The fact that the identity line is not complied with shows that there are systematic patterns of over- or underconfidence. This stratification of these curves by the skin tone and sex reveals subgroup-specific deviations that might be concealed at the dataset level (Thong et al., 2023).

In particular, when the synthetic data augmentation is implemented to enhance the darker-skin lesion cases using GANs or diffusion models (Georges-Filteau and Cirillo, 2020), the domain shift artifacts (Choudhary et al., 2020) may induce subgroup-specific overconfidence in that generated lesions do not display any diagnostically

significant microstructures but still score highly on the malignant classes using the softmax. This can enhance its apparent ROC-AUC but impair calibration, which is a life-threatening illness mismatch in economic-sensitive clinical practice. Quantitative measures of visual reliability curves are provided, including Expected Calibration Error (ECE) and Maximum Calibration Error (MCE), even though a quantitative validation point of view would summarize the amount of deviation. Nonetheless, single scalar errors are to be reported individually by subgroup and globally to ensure transparency in being fair under the GDPR/HIPAA audit criteria (Thong et al., 2023).

A skewed ECE of majority in the global estimation might result in hiding a high minority-group ECE (Wang et al., 2022). By directly including calibration tracking as a monitoring loop within MLOps as described above it is possible to automatically signal ECE or MCE crossing tolerance thresholds within a subgroup, taking remedial measures against affected demographics, both through temperature scaling or mixup-loss adjustment. The semi-supervised learning only aggravates the necessity of regular calibration audits. Pseudo-labels made using weak augmentation methods (FixMatch or MixMatch pipelines) (Yamaguchi, 2023) have confidence levels that cause group inclusion decisions; systematic miscalibration of group-to-group pseudo-labels causes group-dependent acceptance rates of weak pseudo-labels, which compromises both fairness and generalization. The confidence filtering that is provided in RSMatch is inherently tied to the quality of calibration: excessively optimistic scores on low-reliability predictions may introduce numerous samples that are mislabeled into training batches (Wang et al., 2022).

This contamination risk can be minimized at periodic recalibration of logits, either globally or per demographic slice, during SSL cycles at minimal annotation cost overhead cost. Calibration cost-benefit trade-offs are also overlapping with synthetic-to-real integration strategies. Ratio scheduling policies which vary synthetic contribution based on training (Sinha et al., 2024) should include trends of subgroup-calibration with their adjustment signals; increasing sensitivity-at-90percent specificity to Fitzpatrick VI melanomas would not be valuable clinically when the corresponding curves exhibit increasing overconfidence in the benign cases of the same group, in that case such variation would increase false positive burdens disproportionately in that demographic. The artifact/hallucination detection steps described above (Chang et al., 2023) are indirect in that they help to remove implausible synthetics that will distort groupwise calibration with stylistic anomalies that are not based on pathology.

Acquisition variability is another axis that impacts the reliability of probabilities in certain deployment conditions when smartphone-acquired images are used instead of standardized dermoscopy (especially in low-resource clinics), which are defined by the variability of acquisition by images (Kore et al., 2024). Efficient augmentation objectives based on imitating a device-specific perturbation of the light or color balance can minimize miscalibration due to domain-specific effects during inference, however caution is required to avoid chromatic perturbation that violates physiological plausibility between particular Fitzpatrick types (Thong et al., 2023).

The use of device-stratified reliability curves as validation procedures makes sure that per-subgroup alignment of probability is not destroyed during the adaptation steps instead of enhanced. Refinement Calibration refinement algorithms such as isotonic regression, Platt scaling may be used after the fact but these will require held-out validation sets to represent the deployment population at cross-tone and cross-sex scales, which is usually not the case with minority-subgroup samples that are already small. Among the cost-effective solutions to this are weighted calibration fitting in which the underrepresented subgroup errors are upweighted during parameter estimation to achieve parity specifically without requiring unnecessary computer time in full retraining runs (Abir et al., 2024).

In addition to active learning loops that request experts to review cases belonging to miscalibrated bins of important subgroups, annotation budgets focus on those cases that are most likely to decrease clinically relevant miscalibration. It is also important to monitor changes in calibration due to drift even after deployment. As noted above (Kore et al., 2024), neither patient mix nor device infrastructure alteration can bias reliability profiles without presenting any apparent effect on discriminative measures. Integrating subgroup-stratified drift checks to drift dashboards can be discovered early: e.g., an inflow of warmer-colored darker-skin photos moving h3 distribution could be systematically setting back forecasted likelihoods of malignant lesions unless feature space adjusts quickly (Thong et al., 2023).

According to GDPR /HIPAA principles, these snapshots of calibration drifts in combination with model version records serve to provide transparent audit trails earlier in evidence of continued equity assurance. Lastly, clinical credibility is determined not only by eliminating raw performance gaps but also by making probability outputs have similar meaning across demographics at points of operation such as triage thresholds. Fixed probability cutoffs may be used to make decisions regarding proceeding to biopsy, and even equalized sensitivities can result

in inequitable patient outcomes, even when this occurs without the regular inspection of reliability curves on a case-by-case basis (Sagers et al., 2023).

Therefore, the addition of stratified calibration evaluation in cost-performance balancing guarantees that resource-effective methods such as SSL with artificial means of supplementation are beneficial in terms of safety requirements and may honor the pledges of fairness within the current framework of clinical AI governance.

4.2 VALIDATION PROTOCOLS

4.2.1 LOCKED AND EXTERNAL TEST SET USE

Locked and external test set usage plays a central role in ensuring that performance claims that include skin cancer detection tests are clinically credible, cost-effective, and fair across underrepresented demographic strata, for example, higher Fitzpatrick skin tones and for minority sex groups. The difference between the locked and external test is the first and foremost, with regard to their independence from training influences. Locked test sets are built with a limited set of real dermatology images in which the composition of the test set approximates that of the target deployment population without governance on how to ensure that some images will temporarily be incorporated to training or hyperparameter tuning cycles instead (Sagers et al., 2023).

They are kind of "frozen" validation environments which containerize the state of performance at a given point of time without any influence of the data intake thereafter. In fairness-sensitive pipelines such sets that are locked must be stratified demographically, e.g. proportional representation across Fitzpatrick I-VI and balanced sex ratios, for the sub-group specific evaluation of sensitivity at 90 % specificity, ROC-AUC, PR-AUC and calibration curves (Thong et al., 2023).

This is to ensure that headline metrics cannot obscure deteriorations or persistent differences between levels of accuracy in minority groups. External test sets take this principle a step further by introducing data obtained entirely beyond the model development context and ideally from different institutions using different imaging modalities or other devices such as smartphone macro cameras instead of dermatoscopes (Choudhary et al., 2020).

Their purpose is closely related to deployment realism: to meet the need to measure the way well that pipelines generalize to heterogeneous acquisition conditions; keep performance across skin tone strata. For datasets such as ISIC, HAM10000, BCN20000, and Fitzpatrick17k, which share collection biases in favor of lighter skin tones (Pakzad et al., 2022); under-sampled groups smooth fairness groups (e.g., Fitzpatrick IV-VI) are ideally put under-represented in external test sets in order to exercise the fairness motivated augmentation strategies mentioned above. Without such representation in the testing set, methods for synthetic generation, GAN-based conditioning (Georges-Filteau & Cirillo, 2020), diffusion prompt engineering (Tariq et al., 2023) and SSL mixing ratio policies may seem successful on internal evaluations, but not so on real-world demographic compositions. Operationally, both locked and external test sets protect against places where risks of contamination occur to semi-supervised learning workflows. SSL methods such as FixMatch or MixMatch make use of massive unlabeled pools, and these can accidentally consist of near-duplicates or stylistic variables of samples in a test set in case sourcing isn't managed. RSMATCH-style filtering helps on mitigating this problem during the training stage by discarding: low-confidence pseudo-labels (Yamaguchi, 2023), but when designing the protocol, one must design a dataset fingerprinting to avoid training on cases that are visually similar to cases that are reserved to be locked/external tests.

Contamination here is not merely a statistical annoyance, it destroys clinical trust by giving an inflated accuracy which may fail in the case of truly invisible inputs. There is another layer of difficulty to deal with locked/external validation: synthetic/to-real domain shift. A synthetic diversity brings in-training infrequent lesion representations, the advertised fairness advantage, but generated outputs frequently deviate ever so slightly from real lesion structures (Sinha et al., 2024). Locked sets that only contain real clinical images can be used to determine if learned decision boundaries are transferred faithfully outside of artifact patterns that are generated during the synthesis (Chang et al., 2023).

For example, internal ROC-AUC parity between Fitzpatrick II and VI could be eliminated when these tests were externally performed on real type VI lesions taken under non-standard lighting conditions; such degradation could only appear if the test set avoided latent contamination from generating pipelines themselves. Cost-effectiveness crosses at this point through shifting to strategically sized locked/external sets and their reuse policies. Costs of Annotating Large-Scale Dermatology Archives Supersizing the Annotations at Precision of Biopsy Confirmed

Annotations Balancing Cost Pressure Compounds Dermatologist qualities Balance on Protected Diamonds Faulkes Meta-Bioinformatics Patient Identification Experiments (Dhawan and Nijhawan, 2024).

A judicious approach makes use of smaller but demographically-comprehensive held-out splits for continual monitoring while larger external collections are reserved for milestone evaluations accompanied by the revision of major models in registry tracking-systems. Version control is advantageous in the sense that each model artifact should record its performance on both locked and outside benchmarks together with detailed resources spent (such as GPU-hours used during SSL retraining due to drift detection) (Abir et al., 2024).

Artifact / hallucination detection frameworks play indirect but important roles here: a quality assurance applied to synthetic augmentation outputs prior to training helps to reduce spurious cue learning that can depress external test outcomes to minorities despite improvement in internal score (Xing et al, 2023). Embedding these QC results into metadata within a model registry is what allows for post-hoc explanations for any locked/external discrepancy to be traceable whether related to generator degradation (say altered prompt syntax reducing pigment network fidelity) or ratio scheduling shifts leading to changed subgroup exposure dynamics during SSL loops. From the perspective of fairness monitoring, the calibration curves locked/external evaluations highlight whether probability alignment exists outside of the lab conditions (Thong et al., 2023).

Overconfidence drift on minority tones, which are common if one has batches that are dominated by synthetic data in.gov SSL at early epochs, may cause misaligned thresholds in triaging between the internal validation and the real clinic data. Active remediation through temperature scaling directed at each subgroup is only feasible if the nature of miscalibration can be identified through strictly independent testing. Deployment realism goals depend also on external set construction over modality variance: smartphone acquired lesion imagery introduces vast different noise characteristics to dermoscopy datasets such as HAM10000 (Kore et al., 2024).

Including such cases probes MLOps readiness for generalization between devices without hurting the equity, a high ROC-AUC in controlled-learned dermoscopy might conceal poor recall in noisier community-clinic streams without being consciously tested. Long-term governance requires unchangeable storage of locked/external test compositions under access control; GDPR/HIPAA Compliance precludes the unnecessary handling of identifiable patient images but permits aggregate distribution summaries imbedded in registry entries to show demographic coverage in the audit of patients. Drift alerts that happen during live deployments feed back into cycles of re-evaluation on these independent benchmarks: sudden drops in minority group sensitivity-at-90 per cent specificity can be used as the impetus for retrainings based on previous versions that have been validated on all of the slices protected (Wang et al., 2022).

By combining locked/internal parity audits with expansion expanding towards truly external, heterogeneous datasets with a focus on protected attribute coverage, healthcare AI pipelines are granted a defensible foundation to make claims for fairness AND expand its safety case to aid in regulatory compliance. Cost efficiency comes from their size/composition being optimized to balance the need for annotation expense and the desire for statistical power for sensitive subgroup metrics, a balance that can only be achieved through a deliberate protocol embedding within MLOps frameworks that can be made to yield transparent traceability through changing synthetic integration strategies, SSL contamination mitigation stages, and generative source quality controls throughout the lifecycle of clinically deployed models.

5. RISKS AND OPEN ISSUES

5.1 SYNTHETIC-TO-REAL DOMAIN SHIFT

Synthetic to real domain shift occurs when synthetic training data is used to train dermatology AI so using either generative artificial neural networks (GANs), diffusion algorithms or generative-hybrid generative systems when exposed to real clinical data that has different statistical distribution than training data performance are degraded. This divergence is unique and highly consequential considering the ongoing discrepancy between representations found in popular training data sets such as ISIC, HAM10000, BCN20000, and Fitzpatrick17k.19 Pakzad et al. Synthetic generation is often used to fill the gaps, and generate rare lesion presentations in higher Fitzpatrick skin tones or sex minority groups. Nonetheless, if the synthetic images do not approximate the fine-grained diagnostic cue of real lesions, pigment networks, and regression structures, irregular borders, the occurring domain gap can decrease fairness on the level of protected subgroups and jeopardize clinical safety. Texture fidelity is one of the fundamental problems when addressing this shift. Generators trained mostly on lighter-tone lesions induce the reproduction of subtle tonal gradients in those cases that exhibit high quality in visual perception, but induce chromatic or structural artifacts in darker tones (Wang et al, 2022).

These artifacts take the form of, e.g., unnatural color banding, over-smoothing of lesion interiors, implausible transitions in lesion color between lesions and surrounding skin. When such flaws are introduced into synthetic data overrepresented toward tones whereas underrepresented, the learning of the models might prioritize such spurious requirements to the real pathology, which could lead to Civil and accentuates the ROC-AUC in inner validation but the nor in external validation with the aid of real clinical institutional cases (Sagers et al., 2023). This mismatch highlights the observation that augmenting fairness even backfires if the quality of sources is uneven between ambles of demographic slices. Diffusion models have been found to be better at local texture replication than previous GANs (Tariq et al., 2023), which appears to improve the frequency of artifacts and the realism of the outputs of high FST. However, diffusion pipelines are still at risk of introducing the stylistic biases in the pipeline if they are trained on unbalanced base datasets, and the lack of diverse illumination and device-condition reference in the conditioning corpus of a pipeline could lead to narrow output variations that could fail under the realism testing in deployment (Choudhary et al., 2020).

When combining diffusion-generated imagery with semi-supervised learning loops such as FixMatch or MixMatch, misalignment between the synthetic and real may mean making errors for SSL pseudo-label propagation may not be eliminated, especially when confidence thresholds are determined without subgroup conditioning, 2023 by Yamaguchi. RSMATCH-style filtering is useful for reducing the contamination by enforcing multi-view consistency to certify the allowable pseudo-labels; but the performance of this filtering algorithm requires the calibration in the different domains to be equal (Thong et al., 2023), which is a proof itself vulnerable to domain shift. These risks are worsened by differences in device modality. Real-world acquisition conditions are varied from dermoscopic light to smartphone macro, generative models trained solely on dermoscopic data very rarely actually reflect noise patterns with lower resolutions and lower color temperatures than mobile data. This disparity could lead SSL-trained classifiers to fail in low-resource clinics providing higher proportions of their patients with dark skin tones unless target deployment conditions are explicitly simulated through augmentation policies (Kore et al., 2024).

Embedding such simulation into robust aug stages lays the groundwork for making models appropriately prepared for the realistic variability in post-deployment environments without the fact-finding demanding multi-site image gathering; but aggressive simulation must be care taking to not produce funny looking non-physiological tone across specific Fitzpatrick classes. Mixing policies and ratio scheduling has a direct impact on the seriousness of synthetic to real shift during integration into the SSL pipelines (Sinha et al., 2024)

Overweighting the role of synthetic inputs at the beginning of training sets (and thus entrenching style-specific biases) before decision boundaries have been able to incorporate a sufficient amount of exposure to genuine examples from minorities. On the other hand, decreasing synthetic ratios too rapidly can run the risk of losing the benefits of representation before minority equity targets are achieved. Dynamic feedback loops between changes in ratio and subgroup-specific ROC-AUC and calibration drift address both limits and require incessant observation in MLOps frameworks (Pakzad et al., 2022).

Relating these adjustments to artifact/hallucination detection results per subgroup (Chang et al., 2023) guarantees that ratio increases are not deliberately increasing contamination levels in case of reduced generator fidelity. Locked and external test sets are very important tools in uncovering otherwise hidden domain shift effects (Sagers et al., 2023). Internal validation in particular if it involves batches with high synthetic content can highlight

weaknesses that are only widely apparent when testing solely on real-world images taken separately from training data sources (Choudhary et al., 2020).

For fairness-sensitive assessments, these test sets should place a focus on underrepresented demographics in order that drops in performance are revealed where equity interventions are being made most heavily. Discrepancies between locked vs external outcomes are often at root either due to generative source mismatch, or pattern contamination from SSL attributed to unbalanced acceptance rates of pseudo-labels. Cost-efficiency aspects come into play here: minimizing domain shift by doing larger-scale diversifications of synthetic sources or longer cycles of fine-tuning generators requires more computing power and review time from experts. Artifact/hallucination filtering alleviates the wasted effort downstream training but has the cost trade off of itself if thresholds are strict enough to reject half-useful synthetics at scale (Xing et al., 2023).

Optimization of this balance depends on correlation analyses of acceptable counts of synthetics within each subgroup to observed optimizations in the observed metric improvements during external testings to identify diminishing returns points that suggest further generation of synthetics can no longer meet cost performance criteria. Domain adaptation approaches provide an steered response: adversarial feature alignment among real/synthetic divide can help harmonize in latent space geometry before SSL updates are begun (Choudhary et al., 2020), while style transfer pre-processing can be used to get the chromatic profiles of incoming real data more compliant on synthetic baselines, and vice versa, in order to reduce visual gap before the models are ingested.

Yet in each step of adaptation are potential fairness side-effects; overly homogenizing of tonal variations could annihilate clinically relevant pigmentation metrics of certain FSTs; weakening discriminatory specificity in those groups. From a regulatory standpoint from GDPR/HIPAA clauses (Thong et al., 2023) tracing the mitigation processes for a domain shift falls under compliance pardoning, especially well when the representation of protected attributes is tracked explicitly within model registries over version histories (Khader, 2023).

Transparent recording of generator types, conditioning prompts (for tone/sex targeting), augmentation intensities post-generation, and steps to adaptation when used pre-SSL enable auditors to link the outcome in terms of equity with actual design decisions when they consider patient safety claims. Ultimately, the challenge of managing synthetic-to-real domains shift requires coordinated governance of generation quality-control, SSL contamination prevention, ratio-dynamic schedulingPt/competitor coupling through fairness, diverse augmentation simulating domain deployment realism, stringent locked/external test evaluation stratified demo-creation and optimized application of domain adaptation techniques that command clinically salient variation.

5.2 GENERALIZATION LIMITS ACROSS DATASETS

Generalization error between dermatology databases (i.e. ISIC, HAM10000, BCN20000, Fitzpatrick17k): models trained on one data distribution do not generalize well on other datasets with differing demographics, different imaging modalities and/or different lesion-types. The perpetuation of representation bias, asymmetrical coverage, and preference of skin tones corresponding to Fitzpatrick I-III, results in feature extraction bias that is propagated with the model during training through deployment (Pakzad et al., 2022).

Even if fairness-minded augmentation strategies (GAN-based conditioning (Georges-Filteau & Cirillo, 2020), prompt-based diffusion model with skin-tone variability (Tariq et al., 2023)) add minority-tone examples to one corpus, the learned decision boundaries may be focused on corpus-specific stylistic features. These artifacts can be further complicated by acquisition-related set-ups that are dataset dependent, but are visible as a difference in density of pigment network or apparent irregularity of the border at lower tones. Thus, a classifier trained on HAM10000 dermoscopy cases can have high ROC-AUC internally but not prevent sensitivity-at-90% specificity for type V-VI melanomas, as assessed on Fitzpatrick17k smartphone captures, artifacts being fully seen only via stratified external testing (Choudhary et al., 2020).

Threat to credibility: Exacerbated generalisation deficiencies from underrepresented groups detract from clinical credibility, and invalidate fairness benefits assumed from internal validation. These bounds are not being just quantitative but also represent unmodeled covariates not present in the source data: device-specific noise spectra, lighting temperature variations, background skin tone chromaticity in multi-ethnic cohorts different than the dominant style innately represented in ISIC archives. supplementation methods that target the minority classes often worsen this disparity by overfitting between the visual features of the source dataset rather than including more general conditioning variation (Chang et al. 2023).

Presents seamlessly looking dark-skin synthetic lesions obtained from dermoscopic crops bearing lighting consistency adverse to community-clinic smartphone images; these would be of limited use for learning robust tonal invariances from training datasets. Semi-supervised learning pipelines exacerbate these problems when pseudo-labelling is blindly applied over a pool of unlabeled mixed-domain examples (Yamaguchi, 2023).

If model confidence calibration biases towards favorable scores for stylized cases, such as HAM10000-like dermoscopy, the SSL-filter will allow a disproportionate instance of real or synthetic cases that match the style of HAM10000 and reject minority-tone samples that are not confident but likely valuable from the other datasets. The multi-view verification of RSMATCH can reduce the risk of contamination, but cannot completely compensate the effect of calibration difference unless the thresholds of subgroup are dynamically adjusted (Wang et al., 2022).

This interplay also implies that in addition to discriminative metrics (ROC-AUC, PR-AUC), cross-dataset generalization metrics quantify the levels of analysis-wise calibration (PW calibration) for each tuple of modality and protected attribute (Thong et al., 2023). Good test set protocols exercised in controlled (locked) and non-crucial conditions are an important countermeasure to over-optimism. By having standard fixed number of held-out sets per source dataset and completing them with external benchmarks from other institutions/device trainers, evaluator can detect early on those generalization failures (Sagers et al. 2023).

The asymmetry can be observed in the decline of cross-databases performance, which is to be expected for the high-FST cases; this trend indicates the presence of bias even after synthetic addition. External tests that are biased towards underrepresented groups are used to identify whether fairness interventions are transferable across domains or localized improvements that are specific to the generation and augmentation patterns of a single dataset. Prioritized selection of most demographically sparse but clinically relevant subpopulation exemplars helps collecting the highest audit sensitivity under the constrained budget for external brief labeling (Dhawan et al., 2024) and takes resource efficiency bounds while addressing the limited number if applied to a large-scale external set of resources. Domain adaptation is an already proven technical tactic against generalization loss. Such adversarial alignment of the source and target feature spaces neutralizes dataset-specific style features before the identity is impregnated by classification layers as discriminations (Choudhary et al., 2020).

However, the overzealous harmonization can easily lead to the loss of pigmentation information making malignant versus benign lesions undistinguished for some FSTs, directly damaging the fairness goals (Thong et al., 2023). Finally, we apply domain-adaptiveness derived from modality-aware representations to learn to transfer dermoscope-trained features towards smartphone images, maintaining the physiologically relevant distribution of tone variation across signals without averaging. Style-transfer pre-processing is a less aggressive version of adaptation, since it relies on changing only background color properties or exposure light profiles (not great global feature space translation); again, care should be taken to avoid non-physiological tone mapping that lowers clinical interpretability. In addition, augmentation policy tuning has recently been proposed as another way to attain cross-dataset robustness in SSL based methods such as FixMatch or MixMatch when using real plus synthetic sources across domains (Sinha et al., 2024).

Strong augmentation after variance after deployment into the field such as reducing resolution (similar to imperfect optics) enables classifiers to generalize into a wider range of input conditions without re-training the classifier with costly new labeled data. In other works, however, lesion segmentation masks inserted for-lesion augmentation workflows limit transformation intensity, forcing important diagnostic structures to persist intact through robustness training iterations (Georges-Filteau & Cirillo, 2020); uncontrolled augmentation intensity is rapidly burned-out from normal pathology morphology. The occurrence of artifacts and hallucinations varies with generator-dataset combinations and it affects portability of learned representations (Xing et al., 2023).

A GAN fine-tuned on HAM10000 may result in cleaner minority-tone output compared with one trained on more noisy BCN20000 simply due to better image structural prior though this may cause reinforcement of error in SSL loops consuming synthetic bits derived from BCN though cross-dataset check does not detect it. Prior to insertion of QC gating by artifact score, AFL images undergo SSL integration, and only lesion instances that offer up plausible rendering are explored for boundary fine-tuning to encompass the combined, and adventurous, tonal and stylistic distributions of the respective datasets. There is an interactive cost-benefit of broad generator re-training: focused on bridge gaps between datasets, it may use substantial GPU capacity if not matched by commensurate equality of equity increase, due to the limited availability of annotated targets in target domains. By relating optimized ROC-AUC/parity improvement at external benchmarks by compute hours per subgroup-targeted synthesis cycle, teams are able to determine diminishing returns thresholds below which adaptation or augmentation-stepping isn't worth the cost (Khader, 2023).

"These analytics are most useful within MLOps-managed registries where configuration changes are logged against changes of cross source metrics between HAM-sourced synthetics and Fitzpatrick17k reals linking cause (e.g., changed ratio scheduling) to equity-flow outcome over external independent test systems (Pakzad et al., 2022).

Solving generalization limitations: making fairness as a design artifact; learning subianoids across diverse acquisition spaces; subsegmenting pseudo-labels using calibration-aware thresholds; locked/external test regimes; between-fashion borrowing; pathology-preserving masks; synthetic gate absence; fairness cost/ equity performance across truly independent datasets. Without protective measures statistically designed into each iteration of the dermatology AI solutions pipeline, attainable bias diminution within an individual archive will likely dissipate under the stress of the volume and polycharacteristic complexity required in increasingly clinical, multi-source, multi-fidelity deployments.

6. SYNTHESIS AND RECOMMENDATIONS

6.1 INTEGRATED ANSWERS TO DATA GENERATION AND SSL WORKFLOW

Integrating synthetic data synthesis with semi-supervised learning (SSL) for skin cancer detection requires a integration workflow that vacillate th clinical credibility, fairness sensitivity and cost-efficiency without loosing sight of the operational spend in order to operate in a variety of clinical contexts. As discussed above, for annotations such as ISIC, HAM10000, BCN20000, and Fitzpatrick17k, the clinical impact depends on measuring a low annotation cost by correcting underrepresentation in higher Fitzpatrick tones and minority sex cohorts while avoiding deratings due to synthetic-to-real domain shift. This integration extends throughout the generative pipeline design, data curation including quality gating mechanism against artifact/hallucination contamination (Chang et al., 2023) and use of dynamic SSL adaptation mechanisms based on subgroup specific metric driven trends.

On the generation side of things, GAN-based conditional synthesis and diffusion pipelines are complementary since while GANs provide good and fast sample generation with limited resource availability, they might suffer from one or both of issues in mode collapse or tissue colorfulness (Georges-Filteau & Cirillo, 2020); diffusion approaches can synthesize more fine-grained texture parallels and lower artifactousness but come at the cost of high computational costs that should inform the scheduling decision. Conditioning on continuous descriptors (L^* , h^*) of Fitzpatrick categories directly stimulates minority tone (Thong et al. 2023) but should be filtered by patch-level discriminators prey to clinical plausibility while seeking perception realism (Xing et al. 2023).

To avoid uneven quality between demographic slices, and hence lighter tones may be getting cleaner generations than darker ones, audits for artifacts prevalence on subgroup levels are required before SSL is ingested. This provides training signal integrity balance, which is of ultimate importance given fundamental links to downstream performance parity exemplified by ROC-AUC for a fixed specificity (Sagers et al., 2023). Evidently, well-designed mixing policies are critical for the tight coupling with SSL frameworks like FixMatch or MixMatch. The use of synthetic input should be subjected to on/off RSMATCH observances of confidence check multi-view subsidy before receiving pseudolabels (Yamaguchi, 2023).

These checks minimize contamination where model calibration biases result in unequal mislabel acceptance in less represented demographics (a well reported risk with starting points as HAM10000 show a light-skin bias in their dataset - Pakzad et al., 2022). Due to Ratio Scheduling, synthetic contributions are adjusted over training cycles: early synthetic additions maximize minority representation on the decision-bounds; late pruning further reduces exposure to any domain-shift artifacts while active learning highlighted real minority-group coverage (Sinha et al., 2024). As the uncertainty in SSL decays at a predictable pace, active learning loops are naturally connected with minority groups using real examples, which are sent to dermatologists when the pseudo-label agreement in high-FST strata is low (Wang et al., 2022).

As a combination with the generative diversity injection via the prompt engineered synthetics with gap targeting based on the identified gaps on internal fairness audits, this synergy runs to close the representation deficit while reducing the annotation budget. Compute Budgets are consumeable if generative workloads are throttled to increments of ROI quality: Correlation tracking between correlational recurrence Dak sysgen, ROC-AUC insights to synthetic sample throughput is helpful in obtaining cutoffs from whereby more synthetics gain negligible improvement as opposed to cost outlay (Khader, 2023). Deployment realism is enforced using augmentation strategies able to mimic site-to-site capture-condition variability without imposing impact on patient-specific lesion morphology. For example, passing strong augmentations on device noisure patterns who are common in smartphone by macro images as input, prepare models trained in large quality of the data under dermoscrobic enviroments for more questionnable outside test set performances stability (Kore et al., 2024).

Augmentation intensity profiles may be source type dependent, and an increased intensity "clinic simulation" perturbation applied to synthetics early will help close the style gap with real inputs, and then gradually bring the augmentation intensity between real and synthetic samples towards deploying blends. End-to-end checkpoints: locked down test datasets from real-world clinic acquisitions ensure that the fairness gain obtained in the toy lab is externalizable; portability to a holding environment when token populations are not representative by a typical exogenously determined variable (Sagers et al., 2023). Stratified reporting of ROC-AUC, PR-AUC, sensitivity-at-90% specificity, by Fitzpatrick type/sex, helps to ensure that risk assessment is transparent to regulatory audit as a requirement of GDPR/HIPAA (Thong et al., 2023).

Calibration curves also complement these metrics because they show the consistency of the probability alignment, which would have the essential value if outputs are fed directly into threshold select for triage in workflows. MLOps makes these practices into permanent governance tools: the model registry contains links taking you from version to training composition (real/ synthetic mix ratios), GAN/diffusion type and conditional, QC audit results per subgrouping artifact, augmentation policies, ssl filters (Thresh in RSMATCH), and validation rates by protected attributes (Pakzad et al., 2022).

Version control provides support for the functionality of rollback after some fairness measures degrade following changes, for example generator fine-tune improving pigment network realism for dark-toned lesions. Hooks are used to monitor live drift events in input populations or acquisition modalities; automated retraining triggers are used for both ratio or threshold adjustments when subgroup-sensitive sensitivity values fall below predefined tolerances. Even in this integrated setting, it remains a universal problem how to address the synthetic to real domain shift. Adversarial domain adaptation between real/synthetic latents decreases style-specific dependences during classification and does not homogenise tone variations important for pathology detection (Choudhary et al., 2020).

Despite the ubiquitous use of style-transfer processing within the image publishing industry, it must be restricted so that realistic tonality is preserved when travelling from FST to FST so they Kelvin match captured deploys with training synthetics. These changes are introduced downstream of generation, but upstream of the SSL updating process - so feature matching is done before decision-boundary reinforcement starts. Altogether, this unified workflow produces a pipeline of fair-synthetic augmentation-outfed quality-gated SSL-growing-properiation-augmetic policies; deep active learning supercharging minorities-exposure; internal-external validating equity; lineage MLOps logging source-data features, into model parameterizations, into deployment artefacts.

Cost-effectiveness comes not by minimizing cost alone but if expense is explicitly linked to 'hours' of the GPU used for generating or annotating 'minutes' to increase an assessment of piloting 'people' to find facts 'effective' to a 'measurable' 'accurate' reduction in 'demographic' 'gap' without statistical 'masking' at the 'aggregate' level. The system model balances technical viability and clinical legitimacy, as well as ensuring that outputs hold diagnostic validity spanning the range of skin tones and sexes reflected in contemporary dermatology practice, but is auditable under current data governance accreditation regimes.

6.2 PROPOSED RESEARCH AGENDA FOR BIAS MITIGATION AND COST EFFICIENCY.

The research agenda arising out of this review affords experimental underpinning interventions that simultaneously address fairness relationships both within skin tone and sex sub-groups, limit the costs of annotation and computation, and retain deployment realism in differing acquisition environments. Based on the higher level synthetic data-SSL workflow development improvements discussed in Section 6.1 above, the agenda calls for a multi-phase iterative development approach that relies on continuous validation and operational governance. The first axis is dedicated to the development of synthetic data generation methods that can address the protected attribute gaps in the most popular datasets such as ISIC, HAM10000, BCN20000 and Fitzpatrick17k (Pakzad et al., 2022).

Using more matchable ways to provide conditioning which are desperate intensity-based Fitzpatrick types or continuous L-h parameters, conditional GANs are to be compared against diffusion models, conditioned classes of a respective regime (Georges-Filteau & Cirillo, 2020; Tariq et al., 2023). In addition to those generation descriptors, evaluation should stratify artifact prevalence and texture faithfulness at the level of each patch, so as to construct patch-level discriminators in each demographic group, which are combined with the generation quantitative descriptors such as FID/Dist-FID, for guiding generator selection before SSL integration. An explicit aspect of this work will be equalizations for quality balancing strategies, making the congeneric artificial language sounds for majority and minority tones have the same level of plausibility as real asymmetric artifact rates themselves may make a model increasingly biased during training (Wang et al., 2022).

While model training on multiple sites to include device-condition differences is necessary for cross-device generalization in the absence of multi-site real-data collection, these generation experiments can allow device-condition diversity to be embedded into output, etc. In addition to generation fidelity research, development has occurred related to contamination controls in SSL pipelines. FixMatchAgenda estimates subgroup-adaptive confidence thresholds to select between the choices of MixMatch (Yamaguchi, 2023), adjusting to on the fly so

undermined tones are neither exacerbated with a gut-wrenching dearth of inclusion nor drowning in a dyspeptic charge of mislabeled synthetics. Similar to RSMATCH, such multi-view agreement filters would need to be calibrated individually by the errors per subgroup and linearly-calibrated parities based on demographics (Thong et al., 2023). Locking trials should be performed using locked test sets based on real biopsy proven images for each target group (Sagers et al., 2023) in order to be able to identify the effect of the SSL contamination prior to the clinical implementation.

For compute limited "augmentation weak" subsystems, the frequency and strength of augmentations within "augmentation strong" subsystems should be controlled by reducing gain in ROC-AUC per subgroup while training (Khader, 2023). The second major trend is the combination of generative kernels with active learning algorithms. To complement feature-space clustering prior to actual case selection from unlabeled pools, here prompt-engineered synsamples specifically addressing known gaps of representation are used as proxy samples (Thong et al., 2023). Selection policies which use uncertainty measures, entropy or margin sampling; demographically aware diversity metrics such as color-space clustering (in L^*, h^* , for example) to maximize annotation value per case transformed into a full annotation burden by expert annotator(s) with minimal inflation. (Wang et al., 2022).

These strategies should be confirmed in changes of the PR-AUC with changes in ROC-AUC parity to confirm stability of precision for rare malignant lesions in minority tones with the risk of overtriage. Domain shift mitigation still plays an important role on the agenda in light of the synthetic-to-real discrepancies detected above. Adversarial feature matching of real and generated latents will be tested in conjunction with style-transfer preprocessing limited to harmonization of background/tone so as not to wash out pathological pigmentation cues (Choudhary et al., 2020).

The success criteria for this strategy are the external test performance on demographically balanced cohorts without loss of sensitivity-at-90% specificity for any protected group. Thus, external test set methods need to contain an oversampling of historically under-sampled Fitzpatrick types IV-VI lesions acquired under different devices to ensure that the portability of fairness interventions across domain boundaries is truly tested (Sagers et al., 2023). Calibration monitoring is inherently longitudinal, and multiple experiments are maintained for calibration: PMSE.

All augmentation and SSL adaptations will be assessed in terms of subgroup-specific reliability curve and ECE/MCE results and any interventions identified with improvements in ROC/PR but divergence in calibration will lead to customized corrections like adjusting the temperature with weights proportional to magnitude of group miscalibration (Thong et al., 2023). This allows the resulting probability outputs to make sense at triage thresholding after fairness-aware training transformations. At each stage (synthetic generation through SSL update), a central model registry will log demographics info, pass rates of generated samples/artifact/hallucination detection per sub-group (Chang et al., 2023), real/synthetic ratio schedule history, augmentation policy diffs per epoch, calibrated metrics changes per version, cost (GPU-hours/training phase; minutes per annotated dermatologist case).

These data allow retrospective correlation analysis of expenditure directly to equity metric changes; versions showing declining levels of fairness gain per cost are used to inform future budget distribution away from processes with low return on investment. Finally, drift response trends will be developed in MLOps pipelines: N demo-stratified pre- and post-rolling inferences are against locked baselines; if criteria are approached, retraining is done through stable prior versions with corrected ratio scheduling or generator diversity mean shift to the regressed group. Such governance can couple cost-related responsiveness to the ongoing establishment of transparency in compliance with the GDPR/HIPAA requirements based on immutable audit records with clinical usability across heterogeneous environments.

Through these synergetic empirical studies - co-balancing generator fidelity across demographics, co-embedding structural contamination guards in SSL methods, optimally active learning on high-impact case selection strategies, journalism attention general from the conversation dataset/domains under fairness constraints, enforcement calibration parity monitoring, structuring registry-driven governance linked to cost imperatives - the proposed research program addresses the twin impertinent of bias mitigation and cost efficiency as mandated by the imperative of scaling dermatology - AI system into equitable, real-world dermatology summary.

7. CONCLUSION

With advances in synthetic data generation and semi-supervised learning methodologies, new opportunities are emerging to overcome the twisted dynamics in skin cancer detection, that have always remained due to demographic bias in the popularly being used dermatology datasets. By combining the conditional generative models like GANs and diffusion models with adaptive SSL algorithms like FixMatch or MixMatch, it is possible to better represent the higher Fitzpatrick skin types and minority sex groups without the prohibitively expensive costs of annotation. The integration requires close quality controls in case any artifact or hallucination in the synthetic image is created and must be detected and filtered to ensure that all augmentation efforts have added to the generalization of the model rather than the model being biased or noisy.

Dynamic and ratio scheduling policies, which support learning progress by adding artificial examples in a controlled manner, balance the pros and cons of early exposure to underrepresented class subgroups with shortcomings in having enough accurate labeled examples via active learning. This is achieved by targeting computation and expert annotation efforts to the areas where improvement in fairness and accuracy will be the largest. In addition, deployment realism and domain adaptation techniques are used to train models that generalize between heterogeneous clinical environments - including differences in imaging devices, acquisition protocols, etc.

However, thorough validation using locked and external test sets with protected attributes stratum ensures that overfitting and synthetic-to-real domain shift are protected against, and probable discrepancies may otherwise be concealed in aggregate-based metrics. The calibration and reliability analyses also ensure that the predicted probabilities are congruent with the measured probabilities of outcomes across all slices of the demographic index, which is highly important for clinical decision making and patient safety. Continuous struggle on looking at performance drift and unfairness within MLOps frameworks supports immediate detection and resolve of acquire-emerging biases or lack of action form of continued trust beetle throughout the model lifestyles.

A unified research agenda focused on empirical characterisation of generative fidelity, confinement of SSL contamination, active learning efficiency, domain generalisability and calibration parities is needed to translate these technical developments into fair clinical tools. Maintain comprehensive model registries and version control systems in which metadata associated with the demographics, parameters, quality assurance and cost of consideration are clearly recorded for transparent governance and regulatory compliance initiatives (e.g. GDPR, HIPAA) Ultimately, this is an integrated approach where that balances between clinical credibility, fairness sensitivity, and cost-effectiveness, which can provide a viable path to introduce dermatology AI systems that are reliably and equitably functional across diverse patient populations and in real-world consistency.

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